8

BIFURCATIONS REVISITED

8.0 Introduction

This chapter extends our earlier work on bifurcations (Chapter 3). As we move up from one-dimensional to two-dimensional systems, we still find that fixed points can be created or destroyed or destabilized as parameters are varied—but now the same is true of closed orbits as well. Thus we can begin to *describe the ways in which oscillations can be turned on or off.*

In this broader context, what exactly do we mean by a bifurcation? The usual definition involves the concept of "topological equivalence" (Section 6.3): if the phase portrait changes its topological structure as a parameter is varied, we say that a ***bifurcation*** has occurred. Examples include changes in the number or stability of fixed points, closed orbits, or saddle connections as a parameter is varied.

This chapter is organized as follows: for each bifurcation, we start with a simple prototypical example, and then graduate to more challenging examples, either briefly or in separate sections. Models of genetic switches, chemical oscillators, driven pendula and Josephson junctions are used to illustrate the theory.

8.1 Saddle-Node, Transcritical, and Pitchfork Bifurcations

The bifurcations of fixed points discussed in Chapter 3 have analogs in two dimensions (and indeed, in *all* dimensions). Yet it turns out that nothing really new happens when more dimensions are added—all the action is confined to a one-dimensional subspace along which the bifurcations occur, while in the extra dimensions the flow is either simple attraction or repulsion from that subspace, as we'll see below.

Saddle-Node Bifurcation

The saddle-node bifurcation is the basic mechanism for the creation and destruction of fixed points. Here's the prototypical example in two dimensions:

$$\dot{x} = \mu - x^2$$
$$\dot{y} = -y. \qquad (1)$$

In the x-direction we see the bifurcation behavior discussed in Section 3.1, while in the y-direction the motion is exponentially damped.

Consider the phase portrait as μ varies. For $\mu > 0$, Figure 8.1.1 shows that there are two fixed points, a stable node at $(x^*, y^*) = (\sqrt{\mu}, 0)$ and a saddle at $(-\sqrt{\mu}, 0)$. As μ decreases, the saddle and node approach each other, then collide when $\mu = 0$, and finally disappear when $\mu < 0$.

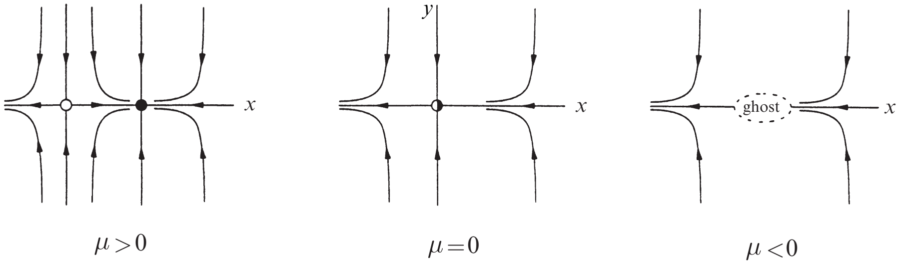

Figure 8.1.1

Even after the fixed points have annihilated each other, they continue to influence the flow—as in Section 4.3, they leave a *ghost*, a bottleneck region that sucks trajectories in and delays them before allowing passage out the other side. For the same reasons as in Section 4.3, the time spent in the bottleneck generically increases as $(\mu - \mu_c)^{-1/2}$, where μ_c is the value at which the saddle-node bifurcation occurs. Some applications of this scaling law in condensed-matter physics are discussed by Strogatz and Westervelt (1989).

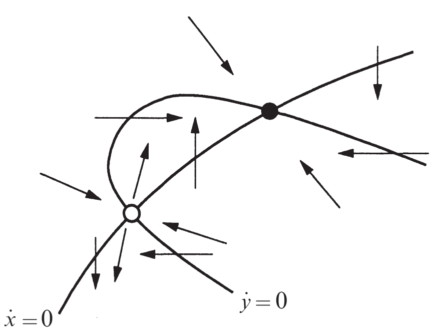

Figure 8.1.2

Figure 8.1.1 is representative of the following more general situation. Consider a two-dimensional system $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$ that depends on a parameter μ. Suppose that for some value of μ the nullclines intersect as shown in Figure 8.1.2. Notice that each intersection corresponds to a fixed point since $\dot{x} = 0$ and $\dot{y} = 0$ simultaneously. Thus, to see how the fixed points move as μ changes, we just have to watch the intersections. Now suppose that the nullclines pull away from each other as μ varies, becoming *tangent* at $\mu = \mu_c$. Then the fixed points approach each other and collide when $\mu = \mu_c$; after the nullclines pull apart, there are no intersections and the fixed points disappear with a bang. The point is that *all* saddle-node bifurcations have this character locally.

EXAMPLE 8.1.1:

The following system has been discussed by Griffith (1971) as a model for a genetic control system. The activity of a certain gene is assumed to be directly induced by two copies of the protein for which it codes. In other words, the gene is stimulated by its own product, potentially leading to an autocatalytic feedback process. In dimensionless form, the equations are

$$\dot{x} = -ax + y$$
$$\dot{y} = \frac{x^2}{1+x^2} - by$$

where x and y are proportional to the concentrations of the protein and the messenger RNA from which it is translated, respectively, and $a, b > 0$ are parameters that govern the rate of degradation of x and y.

Show that the system has three fixed points when $a < a_c$, where a_c is to be determined. Show that two of these fixed points coalesce in a saddle-node bifurcation when $a = a_c$. Then sketch the phase portrait for $a < a_c$, and give a biological interpretation.

Solution: The nullclines are given by the line $y = ax$ and the sigmoidal curve

$$y = \frac{x^2}{b(1+x^2)}$$

as sketched in Figure 8.1.3. Now suppose we vary a while holding b fixed. This is simple to visualize, since a is the slope of the line. For small a there are three intersections, as in Figure 8.1.3. As a increases, the top two intersections approach each other and collide when the line intersects the curve tangentially. For larger values of a, those fixed points disappear, leaving the origin as the only fixed point.

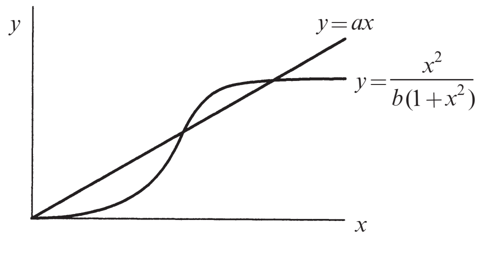

Figure 8.1.3

To find a_c, we compute the fixed points directly and find where they coalesce. The nullclines intersect when

$$ax = \frac{x^2}{b(1+x^2)}.$$

One solution is $x^* = 0$, in which case $y^* = 0$. The other intersections satisfy the quadratic equation

$$ab(1 + x^2) = x \tag{2}$$

which has two solutions

$$x^* = \frac{1 \pm \sqrt{1-4a^2b^2}}{2ab}$$

if $1 - 4a^2 b^2 > 0$, i.e., $2ab < 1$. These solutions coalesce when $2ab = 1$. Hence

$$a_c = 1/2b.$$

For future reference, note that the fixed point $x^* = 1$ at the bifurcation.

The nullclines (Figure 8.1.4) provide a lot of information about the phase portrait for $a < a_c$. The vector field is vertical on the line $y = ax$ and horizontal on the sigmoidal curve. Other arrows can be sketched by noting the signs of $\dot{x}$ and $\dot{y}$. It appears that the middle fixed point is a saddle and the other two are sinks. To confirm this, we turn now to the classification of the fixed points.

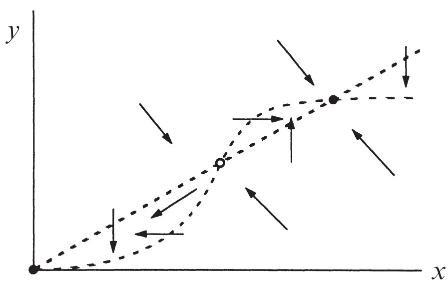

Figure 8.1.4

The Jacobian matrix at (x, y) is

$$A = \begin{pmatrix} -a & 1 \\ \frac{2x}{(1+x^2)^2} & -b \end{pmatrix}.$$

A has trace $\tau = -(a+b) < 0$ so all the fixed points are either sinks or saddles, depending on the value of the determinant Δ. At (0,0), $\Delta = ab > 0$, so the origin is always a stable fixed point. In fact, it is a *stable node*, since $\tau^2 - 4\Delta = (a-b)^2 > 0$ (except in the degenerate case $a = b$, which we disregard). At the other two fixed points, Δ looks messy but it can be simplified using (2). We find

$$\Delta = ab - \frac{2x^*}{\left(1+(x^*)^2\right)^2} = ab\left[1 - \frac{2}{1+(x^*)^2}\right] = ab\left[\frac{(x^*)^2 - 1}{1+(x^*)^2}\right].$$

So $\Delta < 0$ for the "middle" fixed point, which has $0 < x^* < 1$; this is a *saddle point*. The fixed point with $x^* > 1$ is always a *stable node*, since $\Delta < ab$ and therefore $\tau^2 - 4\Delta > (a-b)^2 > 0$.

The phase portrait is plotted in Figure 8.1.5. By looking back at Figure 8.1.4, we can see that the unstable manifold of the saddle is necessarily trapped in the narrow channel between the two nullclines. More importantly, the *stable* manifold separates the plane into two regions, each a basin of attraction for a sink.

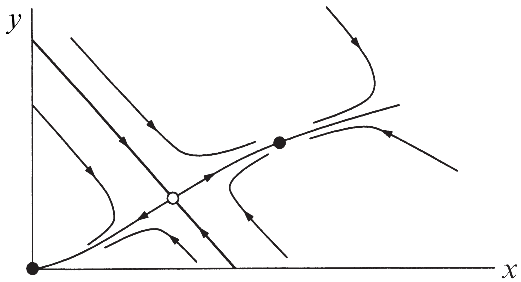

Figure 8.1.5

The biological interpretation is that the system can act like a *biochemical switch*, but only if the mRNA and protein degrade slowly enough—specifically, their decay rates must satisfy $ab < 1/2$. In this case, there are two stable steady states: one at the origin, meaning that the gene is silent and there is no protein around to turn it on; and one where x and y are large, meaning that the gene is active and sustained by the high level of protein. The stable manifold of the saddle acts like a threshold; it determines whether the gene turns on or off, depending on the initial values of x and y. ■

As advertised, the flow in Figure 8.1.5 is qualitatively similar to that in the idealized Figure 8.1.1. All trajectories relax rapidly onto the unstable manifold of the saddle, which plays a completely analogous role to the x-axis in Figure 8.1.1.

Thus, in many respects, the bifurcation is a fundamentally one-dimensional event, with the fixed points sliding toward each other along the unstable manifold like beads on a string. *This is why we spent so much time looking at bifurcations in one-dimensional systems*—they're the building blocks of analogous bifurcations in higher dimensions. (The fundamental role of one-dimensional systems can be justified rigorously by "center manifold theory"—see Wiggins (1990) for an introduction.)

Transcritical and Pitchfork Bifurcations

Using the same idea as above, we can also construct prototypical examples of transcritical and pitchfork bifurcations at a stable fixed point. In the x-direction the dynamics are given by the normal forms discussed in Chapter 3, and in the y-direction the motion is exponentially damped. This yields the following examples:

$$\dot{x} = \mu x - x^2, \qquad \dot{y} = -y \qquad \text{(transcritical)}$$
$$\dot{x} = \mu x - x^3, \qquad \dot{y} = -y \qquad \text{(supercritical pitchfork)}$$
$$\dot{x} = \mu x + x^3, \qquad \dot{y} = -y \qquad \text{(subcritical pitchfork)}$$

The analysis in each case follows the same pattern, so we'll discuss only the supercritical pitchfork, and leave the other two cases as exercises.

EXAMPLE 8.1.2:

Plot the phase portraits for the supercritical pitchfork system $\dot{x} = \mu x - x^3$, $\dot{y} = -y$, for $\mu < 0$, $\mu = 0$, and $\mu > 0$.

Solution: For $\mu < 0$, the only fixed point is a stable node at the origin. For $\mu = 0$, the origin is still stable, but now we have very slow (algebraic) decay along the x-direction instead of exponential decay; this is the phenomenon of "critical slowing down" discussed in Section 3.4 and Exercise 2.4.9. For $\mu > 0$, the origin loses stability and gives birth to two new stable fixed points symmetrically located at $(x^*, y^*) = (\pm\sqrt{\mu}, 0)$. By computing the Jacobian at each point, you can check that the origin is a saddle and the other two fixed points are stable nodes. The phase portraits are shown in Figure 8.1.6. ■

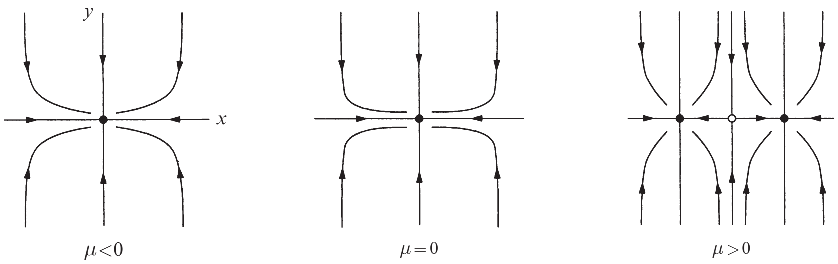

Figure 8.1.6

As mentioned in Chapter 3, pitchfork bifurcations are common in systems that have a symmetry. Here's an example.

EXAMPLE 8.1.3:

Show that a supercritical pitchfork bifurcation occurs at the origin in the system

$$\dot{x} = \mu x + y + \sin x$$
$$\dot{y} = x - y$$

and determine the bifurcation value μ_c. Plot the phase portrait near the origin for μ slightly greater than μ_c.

Solution: The system is invariant under the change of variables $x \to -x$, $y \to -y$, so the phase portrait must be symmetric under reflection through the origin. The origin is a fixed point for all μ, and its Jacobian is

$$A = \begin{pmatrix} \mu+1 & 1 \\ 1 & -1 \end{pmatrix}$$

which has $\tau = \mu$ and $\Delta = -(\mu + 2)$. Hence the origin is a stable fixed point if $\mu < -2$ and a saddle if $\mu > -2$. This suggests that a pitchfork bifurcation occurs at $\mu_c = -2$. To confirm this, we seek a symmetric pair of fixed points close to the origin for μ close to μ_c. (Note that at this stage we don't know whether the bifurcation is sub- or supercritical.) The fixed points satisfy $y = x$ and hence $(\mu + 1)x + \sin x = 0$. One solution is $x = 0$, but we've found that already. Now suppose x is small and nonzero, and expand the sine as a power series. Then

$$(\mu+1)x + x - \frac{x^3}{3!} + O(x^5) = 0.$$

After dividing through by x and neglecting higher-order terms, we get $\mu + 2 - x^2/6 \approx 0$. Hence there is a pair of fixed points with $x^* \approx \pm\sqrt{6(\mu+2)}$ for μ slightly greater than -2. Thus a *supercritical* pitchfork bifurcation occurs at $\mu_c = -2$. (If the bifurcation had been subcritical, the pair of fixed points would exist when the origin was stable, not after it has become a saddle.) Because the bifurcation is supercritical, we know the new fixed points are stable *without even checking.*

To draw the phase portrait near (0,0) for μ slightly greater than -2, it's helpful to find the eigenvectors of the Jacobian at the origin. This can be done exactly, but a simple approximation is that the Jacobian is close to that *at* the bifurcation. Thus

$$A \approx \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$$

which has eigenvectors (1,1) and (1,–1), with eigenvalues $\lambda = 0$ and $\lambda = -2$, respectively. For μ slightly greater than -2 , the origin becomes a saddle and so the zero eigenvalue becomes slightly positive. This information implies the phase portrait shown in Figure 8.1.7.

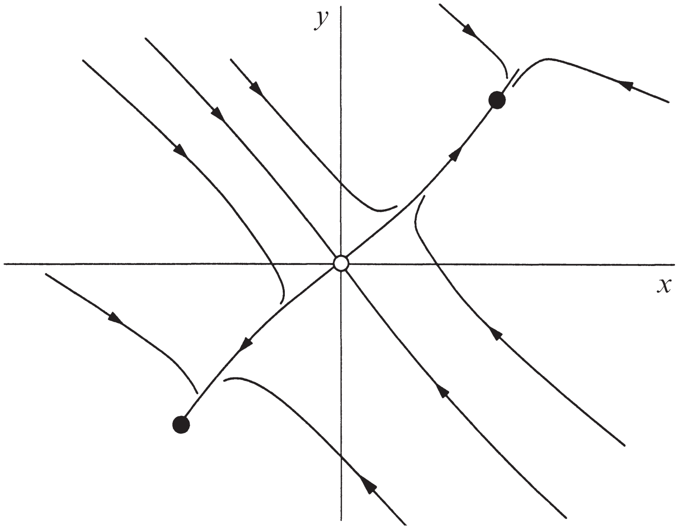

Figure 8.1.7

Note that because of the approximations we've made, this picture is only valid *locally* in both parameter and phase space—if we're not near the origin and if μ is not close to μ_c, all bets are off. ■

In all of the examples above, the bifurcation occurs when $\Delta = 0$, or equivalently, when one of the eigenvalues equals zero. More generally, the saddle-node, transcritical, and pitchfork bifurcations are all examples of ***zero-eigenvalue bifurcations***. (There are other examples, but these are the most common.) Such bifurcations always involve the collision of two or more fixed points.

In the next section we'll consider a fundamentally new kind of bifurcation, one that has no counterpart in one-dimensional systems. It provides a way for a fixed point to lose stability without colliding with any other fixed points.

8.2 Hopf Bifurcations

Suppose a two-dimensional system has a stable fixed point. What are all the possible ways it could lose stability as a parameter μ varies? The eigenvalues of the Jacobian are the key. If the fixed point is stable, the eigenvalues λ_1, λ_2 must both lie in the left half-plane Re $\lambda < 0$. Since the λ's satisfy a quadratic equation with real coefficients, there are two possible pictures: either the eigenvalues are both real and negative (Figure 8.2.1a) or they are complex conjugates (Figure 8.2.1b). To destabilize the fixed point, we need one or both of the eigenvalues to cross into the right half-plane as μ varies.

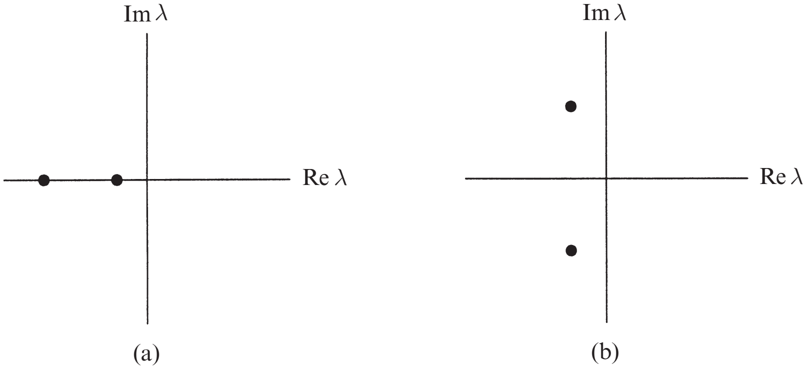

Figure 8.2.1

In Section 8.1 we explored the cases in which a real eigenvalue passes through $\lambda = 0$. These were just our old friends from Chapter 3, namely the saddle-node, transcritical, and pitchfork bifurcations. Now we consider the other possible scenario, in which two complex conjugate eigenvalues simultaneously cross the imaginary axis into the right half-plane.

Supercritical Hopf Bifurcation

Suppose we have a physical system that settles down to equilibrium through exponentially damped oscillations. In other words, small disturbances decay after "ringing" for a while (Figure 8.2.2a). Now suppose that the decay rate depends on a control parameter μ. If the decay becomes slower and slower and finally changes to *growth* at a critical value μ_c, the equilibrium state will lose stability. In many cases the resulting motion is a small-amplitude, sinusoidal, limit cycle oscillation about the former steady state (Figure 8.2.2b). Then we say that the system has undergone a ***supercritical Hopf bifurcation***.

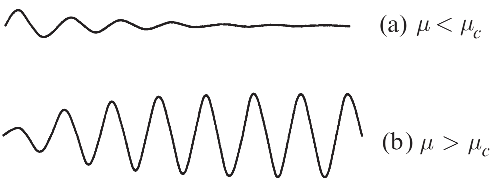

Figure 8.2.2

In terms of the flow in phase space, a supercritical Hopf bifurcation occurs when a stable spiral changes into an unstable spiral surrounded by a small, nearly elliptical limit cycle. Hopf bifurcations can occur in phase spaces of any dimension $n \geq 2$, but as in the rest of this chapter, we'll restrict ourselves to two dimensions.

A simple example of a supercritical Hopf bifurcation is given by the following system:

$$\dot{r} = \mu r - r^3$$
$$\dot{\theta} = \omega + br^2.$$

There are three parameters: μ controls the stability of the fixed point at the origin, ω gives the frequency of infinitesimal oscillations, and b determines the dependence of frequency on amplitude for larger amplitude oscillations.

Figure 8.2.3 plots the phase portraits for μ above and below the bifurcation. When $\mu < 0$ the origin $r = 0$ is a stable spiral whose sense of rotation depends on the sign of ω. For $\mu = 0$ the origin is still a stable spiral, though a very weak one: the decay is only algebraically fast. (This case was shown in Figure 6.3.2. Recall that the linearization wrongly predicts a center at the origin.) Finally, for $\mu > 0$ there is an unstable spiral at the origin and a stable circular limit cycle at $r = \sqrt{\mu}$.

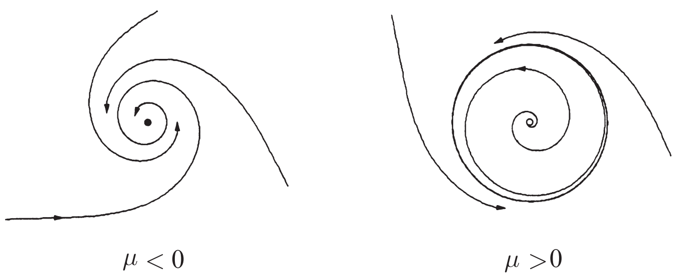

Figure 8.2.3

To see how the eigenvalues behave during the bifurcation, we rewrite the system in Cartesian coordinates; this makes it easier to find the Jacobian. We write $x = r\cos\theta$, $y = r\sin\theta$. Then

$$\begin{aligned}
\dot{x} &= \dot{r}\cos\theta - r\dot{\theta}\sin\theta \\
&= (\mu r - r^3)\cos\theta - r(\omega + br^2)\sin\theta \\
&= \left(\mu - [x^2 + y^2]\right)x - \left(\omega + b[x^2 + y^2]\right)y \\
&= \mu x - \omega y + \text{cubic terms}
\end{aligned}$$

and similarly

$$\dot{y} = \omega x + \mu y + \text{cubic terms}.$$

So the Jacobian at the origin is

$$A = \begin{pmatrix} \mu & -\omega \\ \omega & \mu \end{pmatrix},$$

which has eigenvalues

$$\lambda = \mu \pm i\omega.$$

As expected, the eigenvalues cross the imaginary axis from left to right as μ increases from negative to positive values.

Rules of Thumb

Our idealized case illustrates two rules that hold *generically* for supercritical Hopf bifurcations:

1. The size of the limit cycle grows continuously from zero, and increases proportional to $\sqrt{\mu - \mu_c}$, for μ close to μ_c.
2. The frequency of the limit cycle is given approximately by $\omega = \text{Im } \lambda$, evaluated at $\mu = \mu_c$. This formula is exact at the birth of the limit cycle, and correct within $O(\mu - \mu_c)$ for μ close to μ_c. The period is therefore $T = (2\pi/\text{Im}\,\lambda) + O(\mu - \mu_c)$.

But our idealized example also has some artifactual properties. First, in Hopf bifurcations encountered in practice, the limit cycle is elliptical, not circular, and its shape becomes distorted as μ moves away from the bifurcation point. Our example is only typical topologically, not geometrically. Second, in our idealized case the eigenvalues move on horizontal lines as μ varies, i.e., Im λ is strictly independent of μ. Normally, the eigenvalues would follow a curvy path and cross the imaginary axis with nonzero slope (Figure 8.2.4).

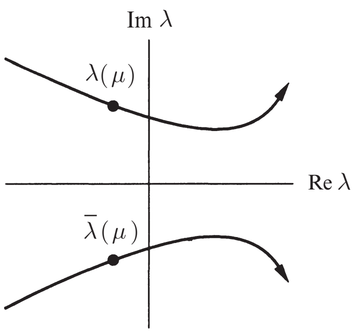

Figure 8.2.4

Subcritical Hopf Bifurcation

Like pitchfork bifurcations, Hopf bifurcations come in both super- and subcritical varieties. The subcritical case is always much more dramatic, and potentially dangerous in engineering applications. After the bifurcation, the trajectories must *jump* to a *distant* attractor, which may be a fixed point, another limit cycle, infinity, or—in three and higher dimensions—a chaotic attractor. We'll see a concrete example of this last, most interesting case when we study the Lorenz equations (Chapter 9).

But for now, consider the two-dimensional example

$$\dot{r} = \mu r + r^3 - r^5$$
$$\dot{\theta} = \omega + br^2.$$

The important difference from the earlier supercritical case is that the cubic term r^3 is now *destabilizing*; it helps to drive trajectories away from the origin.

The phase portraits are shown in Figure 8.2.5. For $\mu < 0$ there are two attractors, a stable limit cycle and a stable fixed point at the origin. Between them lies an unstable cycle, shown as a dashed curve in Figure 8.2.5; it's the player to watch in this scenario. As μ increases, the unstable cycle tightens like a noose around the fixed point. A ***subcritical Hopf bifurcation*** occurs at $\mu = 0$, where the unstable cycle shrinks to zero amplitude and engulfs the origin, rendering it unstable. For $\mu > 0$, the large-amplitude limit cycle is suddenly the only attractor in town. Solutions that used to remain near the origin are now forced to grow into large-amplitude oscillations.

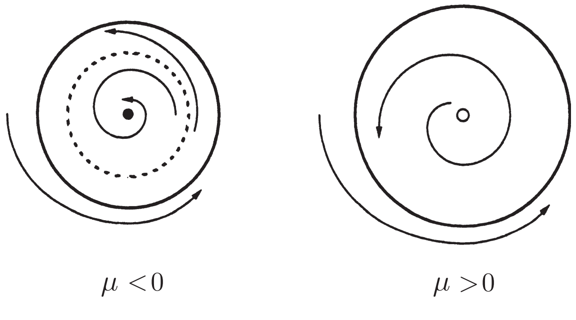

Figure 8.2.5

Note that the system exhibits *hysteresis*: once large-amplitude oscillations have begun, they cannot be turned off by bringing μ back to zero. In fact, the large oscillations will persist until $\mu = -1/4$ where the stable and unstable cycles collide and annihilate. This destruction of the large-amplitude cycle occurs via another type of bifurcation, to be discussed in Section 8.4.

Subcritical Hopf bifurcations occur in the dynamics of nerve cells (Rinzel and Ermentrout 1989), in aeroelastic flutter and other vibrations of airplane wings (Dowell and Ilgamova 1988, Thompson and Stewart 1986), and in instabilities of fluid flows (Drazin and Reid 1981).

Subcritical, Supercritical, or Degenerate Bifurcation?

Given that a Hopf bifurcation occurs, how can we tell if it's sub- or supercritical? The linearization doesn't provide a distinction: in both cases, a pair of eigenvalues moves from the left to the right half-plane.

An analytical criterion exists, but it can be difficult to use (see Exercises 8.2.12–15 for some tractable cases). A quick and dirty approach is to use the computer. If a small, attracting limit cycle appears immediately after the fixed point goes unstable, and if its amplitude shrinks back to zero as the parameter is reversed, the bifurcation is supercritical; otherwise, it's probably subcritical, in which case the nearest attractor might be far from the fixed point, and the system may exhibit hysteresis as the parameter is reversed. Of course, computer experiments are not proofs and you should check the numerics carefully before making any firm conclusions.

Finally, you should also be aware of a ***degenerate Hopf bifurcation***. An example is given by the damped pendulum $\ddot{x}+\mu\dot{x}+\sin x=0$. As we change the damping μ from positive to negative, the fixed point at the origin changes from a stable to an unstable spiral. However at $\mu=0$ we do *not* have a true Hopf bifurcation because there are no limit cycles on either side of the bifurcation. Instead, at $\mu=0$ we have a continuous band of closed orbits surrounding the origin. These are not limit cycles! (Recall that a limit cycle is an *isolated* closed orbit.)

This degenerate case typically arises when a nonconservative system suddenly becomes conservative at the bifurcation point. Then the fixed point becomes a nonlinear center, rather than the weak spiral required by a Hopf bifurcation. See Exercise 8.2.11 for another example.

EXAMPLE 8.2.1:

Consider the system $\dot{x}=\mu x-y+xy^2$, $\dot{y}=x+\mu y+y^3$. Show that a Hopf bifurcation occurs at the origin as μ varies. Is the bifurcation subcritical, supercritical, or degenerate?

Solution: The Jacobian at the origin is $A=\begin{pmatrix}\mu & -1\\ 1 & \mu\end{pmatrix}$, which has $\tau = 2\mu$, $\Delta = \mu^2 + 1 > 0$, and $\lambda = \mu \pm i$. Hence, as μ increases through zero, the origin changes from a stable spiral to an unstable spiral. This suggests that some kind of Hopf bifurcation takes place at $\mu=0$.

To decide whether the bifurcation is subcritical, supercritical, or degenerate, we use simple reasoning and numerical integration. If we transform the system to polar coordinates, we find that

$$\dot{r} = \mu r + r y^2,$$

as you should check. Hence $\dot{r} \geq \mu r$. This implies that for $\mu > 0$, $r(t)$ grows *at least* as fast as $r_0 e^{\mu t}$. In other words, all trajectories are repelled out to infinity! So there are certainly no closed orbits for $\mu > 0$. In particular, the unstable spiral is not surrounded by a stable limit cycle; hence the bifurcation cannot be supercritical.

Could the bifurcation be degenerate? That would require that the origin be a nonlinear center when $\mu = 0$. But $\dot{r}$ is strictly positive away from the x-axis, so closed orbits are still impossible.

By process of elimination, we expect that the bifurcation is *subcritical.* This is confirmed by Figure 8.2.6, which is a computer-generated phase portrait for $\mu = -0.2$.

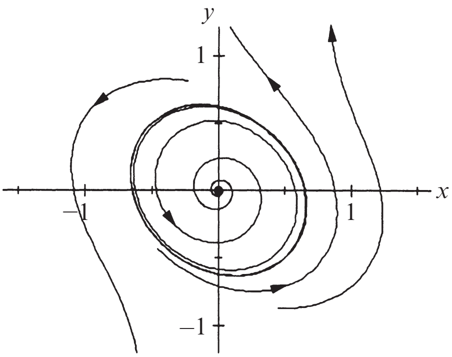

Figure 8.2.6

Note that an *unstable* limit cycle surrounds the stable fixed point, just as we expect in a subcritical bifurcation. Furthermore, the cycle is nearly elliptical and surrounds a gently winding spiral—these are typical features of *either* kind of Hopf bifurcation. ■

8.3 Oscillating Chemical Reactions

For an application of Hopf bifurcations, we now consider a class of experimental systems known as ***chemical oscillators***. These systems are remarkable, both for their spectacular behavior and for the story behind their discovery. After presenting this background information, we analyze a simple model for oscillations in the chlorine dioxide–iodine–malonic acid reaction. The definitive reference on chemical oscillations is the book edited by Field and Burger (1985). See also Epstein et al. (1983), Winfree (1987b) and Murray (2002).

Belousov's "Supposedly Discovered Discovery"

In the early 1950s the Russian biochemist Boris Belousov was trying to create a test tube caricature of the Krebs cycle, a metabolic process that occurs in living cells. When he mixed citric acid and bromate ions in a solution of sulfuric acid, and in the presence of a cerium catalyst, he observed to his astonishment that the mixture became yellow, then faded to colorless after about a minute, then returned to yellow a minute later, then became colorless again, and continued to oscillate dozens of times before finally reaching equilibrium after about an hour.

Today it comes as no surprise that chemical reactions can oscillate spontaneously—such reactions have become a standard demonstration in chemistry classes, and you may have seen one yourself. (For recipes, see Winfree (1980).) But in Belousov's day, his discovery was so radical that he couldn't get his work published. It was thought that all solutions of chemical reagents must go *monotonically* to equilibrium, because of the laws of thermodynamics. Belousov's paper was rejected by one journal after another. According to Winfree (1987b, p.161), one editor even added a snide remark about Belousov's "supposedly discovered discovery" to the rejection letter.

Belousov finally managed to publish a brief abstract in the obscure proceedings of a Russian medical meeting (Belousov 1959), although his colleagues weren't aware of it until years later. Nevertheless, word of his amazing reaction circulated among Moscow chemists in the late 1950s, and in 1961 a graduate student named Zhabotinsky was assigned by his adviser to look into it. Zhabotinsky confirmed that Belousov was right all along, and brought this work to light at an international conference in Prague in 1968, one of the few times that Western and Soviet scientists were allowed to meet. At that time there was a great deal of interest in biological and biochemical oscillations (Chance et al. 1973) and the BZ reaction, as it came to be called, was seen as a manageable model of those more complex systems.

The analogy to biology turned out to be surprisingly close: Zaikin and Zhabotinsky (1970) and Winfree (1972) observed beautiful propagating *waves* of oxidation in thin unstirred layers of BZ reagent, and found that these waves annihilate upon collision, just like waves of excitation in neural or cardiac tissue. The waves always take the shape of expanding concentric rings or spirals (Color plate 1). Spiral waves are now recognized to be a ubiquitous feature of chemical, biological, and physical excitable media; in particular, spiral waves and their three-dimensional analogs, "scroll waves", appear to be implicated in certain cardiac arrhythmias, a problem of great medical importance (Winfree 1987b).

Boris Belousov would be pleased to see what he started.

In 1980, he and Zhabotinsky were awarded the Lenin Prize, the Soviet Union's highest medal, for their pioneering work on oscillating reactions. Unfortunately, Belousov had passed away ten years earlier.

For more about the history of the BZ reaction, see Winfree (1984, 1987b). An English translation of Belousov's original paper from 1951 appears in Field and Burger (1985).

Chlorine Dioxide–Iodine–Malonic Acid Reaction

The mechanisms of chemical oscillations can be very complex. The BZ reaction is thought to involve more than twenty elementary reaction steps, but luckily many of them equilibrate rapidly—this allows the kinetics to be reduced to as few as three differential equations. See Tyson (1985) for this reduced system and its analysis.

In a similar spirit, Lengyel et al. (1990) have proposed and analyzed a particularly elegant model of another oscillating reaction, the chlorine dioxide-iodine-malonic acid ($C1O_2$-I_2-MA) reaction. Their experiments show that the following three reactions and empirical rate laws capture the behavior of the system:

$$\mathrm{MA}+\mathrm{I}_2 \to \mathrm{IMA}+\mathrm{I}^- +\mathrm{H}^+; \quad \frac{d[\mathrm{I}_2]}{dt} = -\frac{k_{1a}[\mathrm{MA}][\mathrm{I}_2]}{k_{1b}+[\mathrm{I}_2]} \tag{1}$$

$$\mathrm{ClO}_2 + \mathrm{I}^- \to \mathrm{ClO}_2^{\,-} + \tfrac{1}{2}\mathrm{I}_2; \quad \frac{d[\mathrm{ClO}_2]}{dt} = -k_2[\mathrm{ClO}_2][\mathrm{I}^-] \tag{2}$$

$$\mathrm{ClO}_2^{\,-} + 4\mathrm{I}^- + 4\mathrm{H}^+ \to \mathrm{Cl}^- + 2\mathrm{I}_2 + 2\mathrm{H}_2\mathrm{O};$$

$$\frac{d[\mathrm{ClO}_2^{\,-}]}{dt} = -k_{3a}[\mathrm{ClO}_2^{\,-}][\mathrm{I}^-][\mathrm{H}^+] - k_{3b}[\mathrm{ClO}_2^{\,-}][\mathrm{I}_2]\frac{[\mathrm{I}^-]}{u+[\mathrm{I}^-]^2} \tag{3}$$

Typical values of the concentrations and kinetic parameters are given in Lengyel et al. (1990) and Lengyel and Epstein (1991).

Numerical integrations of (l)–(3) show that the model exhibits oscillations that closely resemble those observed experimentally. However this model is still too complicated to handle analytically. To simplify it, Lengyel et al. (1990) use a result found in their simulations: Three of the reactants (MA, I_2, and ClO_2) vary much more slowly than the intermediates I^- and $C1O_2^-$, which change by several orders of magnitude during an oscillation period. By approximating the concentrations of the slow reactants as *constants* and making other reasonable simplifications, they reduce the system to a two-variable model. (Of course, since this approximation neglects the slow consumption of the reactants, the model will be unable to account for the eventual approach to equilibrium.) After suitable nondimensionalization, the model becomes

$$\dot{x} = a - x - \frac{4xy}{1+x^2} \qquad (4)$$

$$\dot{y} = bx\left(1 - \frac{y}{1+x^2}\right) \qquad (5)$$

where x and y are the dimensionless concentrations of I^- and ClO_2^-. The parameters a, $b > 0$ depend on the empirical rate constants and on the concentrations assumed for the slow reactants.

We begin the analysis of (4), (5) by constructing a trapping region and applying the Poincaré–Bendixson theorem. Then we'll show that the chemical oscillations arise from a supercritical Hopf bifurcation.

EXAMPLE 8.3.1:

Prove that the system (4), (5) has a closed orbit in the positive quadrant $x, y > 0$ if a and b satisfy certain constraints, to be determined.

Solution: As in Example 7.3.2, the nullclines help us to construct a trapping region. Equation (4) shows that $\dot{x} = 0$ on the curve

$$y = \frac{(a-x)(1+x^2)}{4x} \qquad (6)$$

and (5) shows that $\dot{y} = 0$ on the y-axis and on the parabola $y = 1 + x^2$. These nullclines are sketched in Figure 8.3.1, along with some representative vectors.

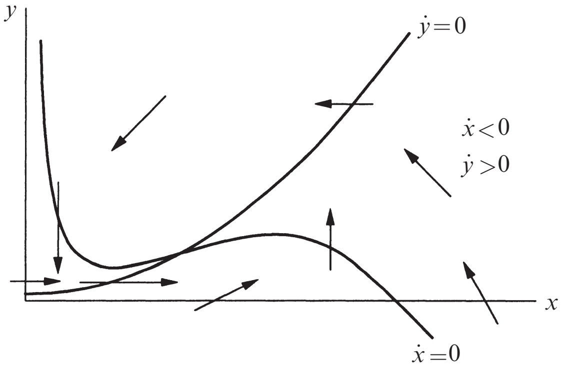

Figure 8.3.1

(We've taken some pedagogical license with Figure 8.3.1; the curvature of the nullcline (6) has been exaggerated to highlight its shape, and to give us more room to draw the vectors.)

Now consider the dashed box shown in Figure 8.3.2. It's a trapping region because all the vectors on the boundary point into the box.

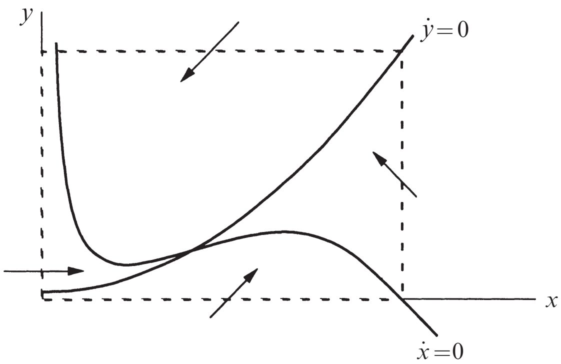

Figure 8.3.2

We can't apply the Poincaré–Bendixson theorem yet, because there's a fixed point

$$x^* = a/5, \qquad y^* = 1+(x^*)^2 = 1+(a/5)^2$$

inside the box at the intersection of the nullclines. But now we argue as in Example 7.3.3: if the fixed point turns out to be a *repeller*, we *can* apply the Poincaré-Bendixson theorem to the "punctured" box obtained by removing the fixed point.

All that remains is to see under what conditions (if any) the fixed point is a repeller. The Jacobian at (x^*, y^*) is

$$\frac{1}{1+(x^*)^2}\begin{pmatrix} 3(x^*)^2-5 & -4x^* \\ 2b(x^*)^2 & -bx^* \end{pmatrix}.$$

(We've used the relation $y^* = 1 + (x^*)^2$ to simplify some of the entries in the Jacobian.) The determinant and trace are given by

$$\Delta = \frac{5bx^*}{1+(x^*)^2} > 0, \qquad \tau = \frac{3(x^*)^2-5-bx^*}{1+(x^*)^2}.$$

We're in luck—since $\Delta > 0$, the fixed point is never a saddle. Hence (x^*, y^*) is a repeller if $\tau > 0$, i.e., if

$$b < b_c \equiv 3a/5 - 25/a. \qquad (7)$$

When (7) holds, the Poincaré-Bendixson theorem implies the existence of a closed orbit somewhere in the punctured box. ■

EXAMPLE 8.3.2:

Using numerical integration, show that a Hopf bifurcation occurs at $b = b_c$ and decide whether the bifurcation is sub- or supercritical.

Solution: The analytical results above show that as b decreases through b_c, the fixed point changes from a stable spiral to an unstable spiral; this is the signature of a Hopf bifurcation. Figure 8.3.3 plots two typical phase portraits. (Here we have chosen $a = 10$; then (7) implies $b_c = 3.5$.) When $b > b_c$, all trajectories spiral into the stable fixed point (Figure 8.3.3a), while for $b < b_c$ they are attracted to a stable limit cycle (Figure 8.3.3b).

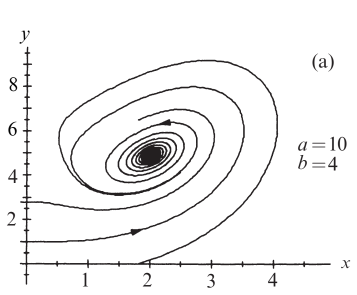

Figure 8.3.3

Hence the bifurcation is *supercritical*—after the fixed point loses stability, it is surrounded by a stable limit cycle. Moreover, by plotting phase portraits as $b \to b_c$ from below, we could confirm that the limit cycle shrinks continuously to a point, as required. ■

Our results are summarized in the stability diagram in Figure 8.3.4. The boundary between the two regions is given by the Hopf bifurcation locus $b = 3a/5–25/a$.

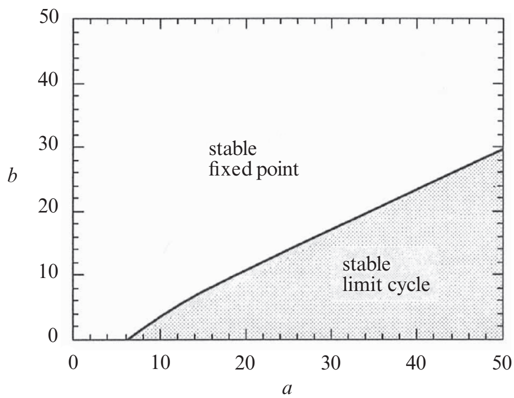

Figure 8.3.4

EXAMPLE 8.3.3:

Approximate the period of the limit cycle for b slightly less than b_c.

Solution: The frequency is approximated by the imaginary part of the eigenvalues at the bifurcation. As usual, the eigenvalues satisfy $\lambda^2 - \tau\lambda + \Delta = 0$. Since $\tau = 0$ and $\Delta > 0$ at $b = b_c$, we find

$$\lambda = \pm i\sqrt{\Delta}.$$

But at b_c,

$$\Delta = \frac{5b_c x^*}{1+(x^*)^2} = \frac{5\left(\frac{3a}{5} - \frac{25}{a}\right)\left(\frac{a}{5}\right)}{1+(a/5)^2} = \frac{15a^2 - 625}{a^2 + 25}.$$

Hence $\omega \approx \Delta^{1/2} = \left[(15a^2 - 625)/(a^2+25)\right]^{1/2}$ and therefore

$$\begin{aligned} T &= 2\pi/\omega \\ &= 2\pi\left[(a^2+25)/(15a^2-625)\right]^{1/2}. \end{aligned}$$

A graph of $T(a)$ is shown in Figure 8.3.5. As $a \to \infty$, $T \to 2\pi/\sqrt{15} \approx 1.63$. ■

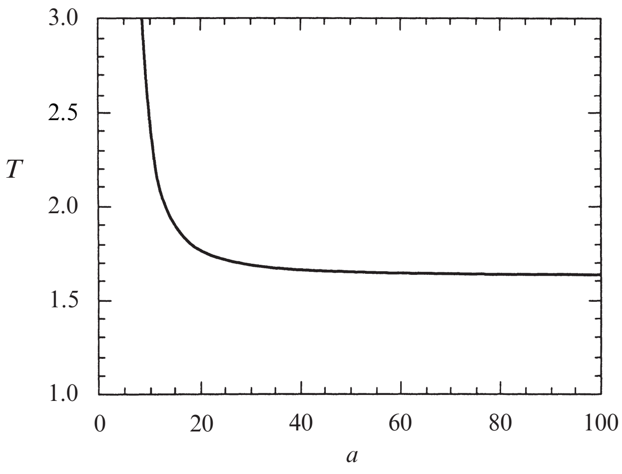

Figure 8.3.5

8.4 Global Bifurcations of Cycles

In two-dimensional systems, there are four common ways in which limit cycles are created or destroyed. The Hopf bifurcation is the most famous, but the other three deserve their day in the sun. They are harder to detect because they involve large regions of the phase plane rather than just the neighborhood of a single fixed point. Hence they are called ***global bifurcations***. In this section we offer some prototypical examples of global bifurcations, and then compare them to one another and to the Hopf bifurcation. A few of their scientific applications are discussed in Sections 8.5 and 8.6 and in the exercises.

Saddle-node Bifurcation of Cycles

A bifurcation in which two limit cycles coalesce and annihilate is called a *fold* or ***saddle-node bifurcation of cycles***, by analogy with the related bifurcation of fixed points. An example occurs in the system

$$\dot{r} = \mu r + r^3 - r^5$$
$$\dot{\theta} = \omega + br^2$$

studied in Section 8.2. There we were interested in the subcritical Hopf bifurcation at $\mu = 0$; now we concentrate on the dynamics for $\mu < 0$.

It is helpful to regard the radial equation $\dot{r} = \mu r + r^3 - r^5$ as a one-dimensional system. As you should check, this system undergoes a saddle-node bifurcation of fixed points at $\mu_c = -1/4$. Now returning to the two-dimensional system, these fixed points correspond to circular *limit cycles*. Figure 8.4.1 plots the "radial phase portraits" and the corresponding behavior in the phase plane.

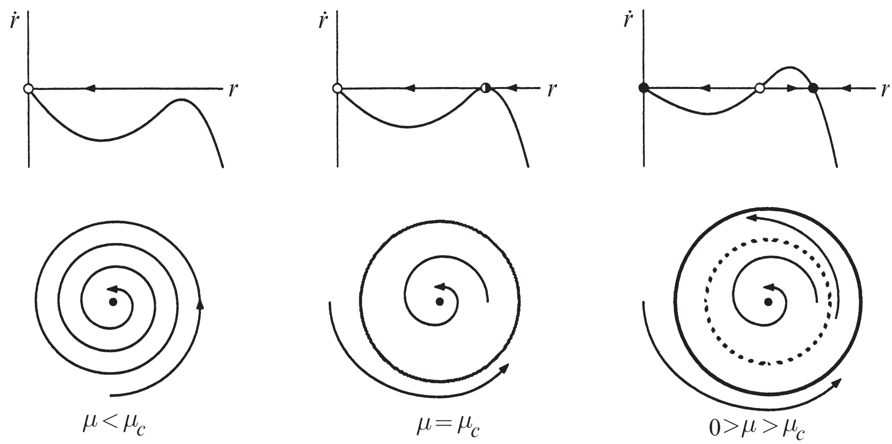

Figure 8.4.1

At μ_c a half-stable cycle is born out of the clear blue sky. As μ increases it splits into a pair of limit cycles, one stable, one unstable. Viewed in the other direction, a stable and unstable cycle collide and disappear as μ decreases through μ_c. Notice that the origin remains stable throughout; it does not participate in this bifurcation.

For future reference, note that at birth the cycle has $O(1)$ amplitude, in contrast to the Hopf bifurcation, where the limit cycle has small amplitude proportional to $(\mu - \mu_c)^{1/2}$.

Infinite-period Bifurcation

Consider the system

$$\dot{r} = r(1 - r^2)$$
$$\dot{\theta} = \mu - \sin\theta$$

where $\mu \geq 0$. This system combines two one-dimensional systems that we have studied previously in Chapters 3 and 4. In the radial direction, all trajectories (except $r^* = 0$) approach the unit circle monotonically as $t \to \infty$. In the angular direction, the motion is everywhere counterclockwise if $\mu > 1$, whereas there are two invariant rays defined by $\sin\theta = \mu$ if $\mu < 1$. Hence as μ decreases through $\mu_c = 1$, the phase portraits change as in Figure 8.4.2.

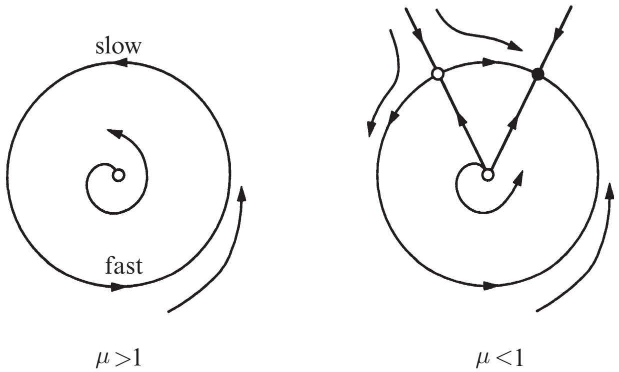

Figure 8.4.2

As μ decreases, the limit cycle $r = 1$ develops a bottleneck at $\theta = \pi/2$ that becomes increasingly severe as $\mu \to 1^+$. The oscillation period lengthens and finally becomes infinite at $\mu_c = 1$, when a fixed point appears on the circle; hence the term ***infinite-period bifurcation***. For $\mu < 1$, the fixed point splits into a saddle and a node.

As the bifurcation is approached, the amplitude of the oscillation stays $O(1)$ but the period increases like $(\mu - \mu_c)^{-1/2}$, for the reasons discussed in Section 4.3.

Homoclinic Bifurcation

In this scenario, part of a limit cycle moves closer and closer to a saddle point. At the bifurcation the cycle touches the saddle point and becomes a homoclinic orbit. This is another kind of infinite-period bifurcation; to avoid confusion, we'll call it a *saddle-loop* or ***homoclinic bifurcation***.

It is hard to find an analytically transparent example, so we resort to the computer. Consider the system

$$\dot{x} = y$$
$$\dot{y} = \mu y + x - x^2 + xy.$$

Figure 8.4.3 plots a series of phase portraits before, during, and after the bifurcation; only the important features are shown.

Numerically, the bifurcation is found to occur at $\mu_c \approx -0.8645$. For $\mu < \mu_c$, say $\mu = -0.92$, a stable limit cycle passes close to a saddle point at the origin (Figure 8.4.3a). As μ increases to μ_c, the limit cycle swells (Figure 8.4.3b) and bangs into the saddle, creating a homoclinic orbit (Figure 8.4.3c). Once $\mu > \mu_c$, the saddle connection breaks and the loop is destroyed (Figure 8.4.3d).

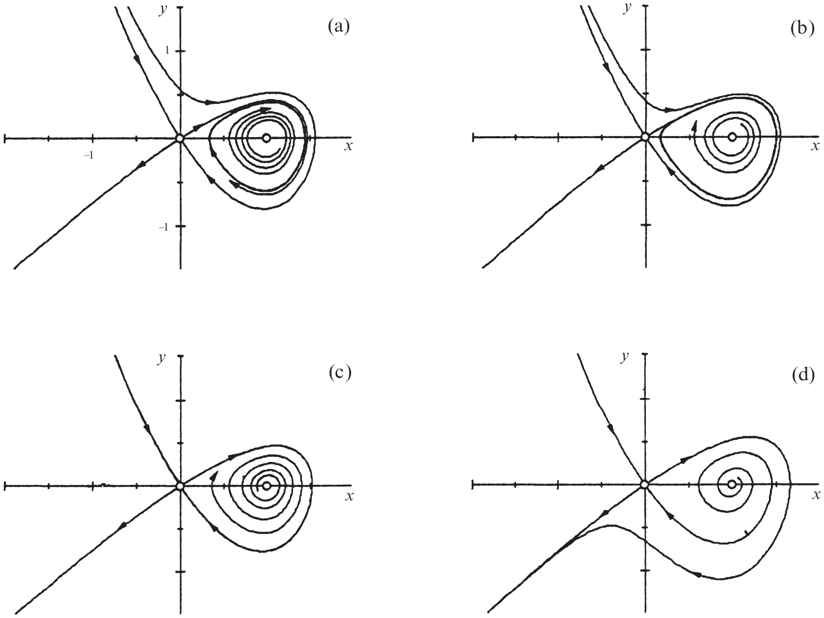

Figure 8.4.3

The key to this bifurcation is the behavior of the unstable manifold of the saddle. Look at the branch of the unstable manifold that leaves the origin to the northeast: after it loops around, it either hits the origin (Figure 8.4.3c) or veers off to one side or the other (Figures 8.4.3a, d).

Scaling Laws

For each of the bifurcations given here, there are characteristic *scaling laws* that govern the amplitude and period of the limit cycle as the bifurcation is approached. Let μ denote some dimensionless measure of the distance from the bifurcation, and assume that $\mu << 1$. The generic scaling laws for bifurcations of cycles in two-dimensional systems are given in Table 7.4.1.

	Amplitude of stable limit cycle	Period of cycle
Supercritical Hopf	$O(\mu^{1/2})$	$O(1)$
Saddle-node bifurcation of cycles	$O(1)$	$O(1)$
Infinite-period	$O(1)$	$O(\mu^{-1/2})$
Homoclinic	$O(1)$	$O(\ln \mu)$

Table 8.4.1

All of these laws have been explained previously, except those for the homoclinic bifurcation. The scaling of the period in that case is obtained by estimating the time required for a trajectory to pass by a saddle point (see Exercise 8.4.12 and Gaspard 1990).

Exceptions to these rules can occur, but only if there is some symmetry or other special feature that renders the problem nongeneric, as in the following example.

EXAMPLE 8.4.1:

The van der Pol oscillator $\ddot{x} + \varepsilon\dot{x}(x^2 - 1) + x = 0$ does not seem to fit anywhere in Table 7.4.1. At $\varepsilon = 0$, the eigenvalues at the origin are pure imaginary ($\lambda = \pm i$), suggesting that a Hopf bifurcation occurs at $\varepsilon = 0$. But we know from Section 7.6 that for $0 < \varepsilon << 1$, the system has a limit cycle of amplitude $r \approx 2$. Thus the cycle is born "full grown," not with size $O(\varepsilon^{1/2})$ as predicted by the scaling law. What's the explanation?

Solution: The bifurcation at $\varepsilon = 0$ is degenerate. The nonlinear term $\varepsilon\dot{x}x^2$ vanishes at precisely the same parameter value as the eigenvalues cross the imaginary axis. That's a nongeneric coincidence if there ever was one!

We can rescale x to remove this degeneracy. Write the equation as $\ddot{x} + x + \varepsilon x^2\dot{x} - \varepsilon\dot{x} = 0$. Let $u^2 = \varepsilon\, x^2$ to remove the ε-dependence of the nonlinear term. Then $u = \varepsilon^{1/2}x$ and the equation becomes

$$\ddot{u} + u + u^2\dot{u} - \varepsilon\dot{u} = 0.$$

Now the nonlinear term is not destroyed when the eigenvalues become pure imaginary. From Section 7.6 the limit cycle solution is $x(t,\varepsilon) \approx 2\cos t$ for $0 < \varepsilon << 1$. In terms of u this becomes

$$u(t,\varepsilon) \approx \left(2\sqrt{\varepsilon}\right)\cos t.$$

Hence the amplitude grows like $\varepsilon^{1/2}$, just as expected for a Hopf bifurcation. ■

The scaling laws given here were derived by thinking about prototypical examples in *two-dimensional* systems. In higher-dimensional phase spaces, the corresponding bifurcations obey the same scaling laws, but with two caveats: (1) Many *additional* bifurcations of limit cycles become possible; thus our table is no longer exhaustive. (2) The homoclinic bifurcation becomes much more subtle to analyze. It often creates chaotic dynamics in its aftermath (Guckenheimer and Holmes 1983, Wiggins 1990).

All of this begs the question: Why should you care about these scaling laws? Suppose you're an experimental scientist and the system you're studying exhibits a stable limit cycle oscillation. Now suppose you change a control parameter and the oscillation stops. By examining the scaling of the period and amplitude near this bifurcation, you can learn something about the system's dynamics (which are usually not known precisely, if at all). In this way, possible models can be eliminated or supported. For an example in physical chemistry, see Gaspard (1990).

8.5 Hysteresis in the Driven Pendulum and Josephson Junction

This section deals with a physical problem in which both homoclinic and infinite-period bifurcations arise. The problem was introduced back in Sections 4.4 and 4.6. At that time we were studying the dynamics of a damped pendulum driven by a constant torque, or equivalently, its high-tech analog, a superconducting Josephson junction driven by a constant current. Because we weren't ready for two-dimensional systems, we reduced both problems to vector fields on the circle by looking at the heavily *overdamped limit* of negligible mass (for the pendulum) or negligible capacitance (for the Josephson junction).

Now we're ready to tackle the full two-dimensional problem. As we claimed at the end of Section 4.6, for sufficiently weak damping the pendulum and the Josephson junction can exhibit intriguing hysteresis effects, thanks to the coexistence of a stable limit cycle and a stable fixed point. In physical terms, the pendulum can settle into either a rotating solution where it whirls over the top, or a stable rest state where gravity balances the applied torque. The final state depends on the initial conditions. Our goal now is to understand how this bistability comes about.

We will phrase our discussion in terms of the Josephson junction, but will mention the pendulum analog whenever it seems helpful.

Governing Equations

As explained in Section 4.6, the governing equation for the Josephson junction is

$$\frac{\hbar C}{2e}\ddot{\phi}+\frac{\hbar}{2eR}\dot{\phi}+I_c\sin\phi=I_B \tag{1}$$

where $\hbar$ is Planck's constant divided by 2π, e is the charge on the electron, I_B is the constant bias current, C, R, and I_c are the junction's capacitance, resistance, and critical current, and $\phi(t)$ is the phase difference across the junction.

To highlight the role of damping, we nondimensionalize (1) differently from in Section 4.6. Let

$$\tilde{t}=\left(\frac{2eI_c}{\hbar C}\right)^{1/2}t,\qquad I=\frac{I_B}{I_c},\qquad \alpha=\left(\frac{\hbar}{2eI_cR^2C}\right)^{1/2}. \tag{2}$$

Then (1) becomes

$$\phi''+\alpha\phi'+\sin\phi=I \tag{3}$$

where α and I are the dimensionless damping and applied current, and the prime denotes differentiation with respect to $\tilde{t}$. Here $\alpha>0$ on physical grounds, and we may choose $I\geq 0$ without loss of generality (otherwise, redefine $\phi\to-\phi$).

Let $y=\phi'$. Then the system becomes

$$\begin{aligned}\phi'&=y\\ y'&=I-\sin\phi-\alpha y.\end{aligned} \tag{4}$$

As in Section 6.7 the phase space is a *cylinder*, since ϕ is an angular variable and y is a real number (best thought of as an angular velocity).

Fixed Points

The fixed points of (4) satisfy $y^*=0$ and $\sin\phi^*=I$. Hence there are two fixed points on the cylinder if $I<1$, and none if $I>1$. When the fixed points exist, one is a saddle and other is a sink, since the Jacobian

$$A=\begin{pmatrix}0 & 1\\ -\cos\phi^* & -\alpha\end{pmatrix}$$

has $\tau = -\alpha < 0$ and $\Delta = \cos\phi^* = \pm\sqrt{1-I^2}$. When $\Delta > 0$, we have a stable node if $\tau^2 - 4\Delta = \alpha^2 - 4\sqrt{1-I^2} > 0$, i.e., if the damping is strong enough or if I is close to 1; otherwise the sink is a stable spiral. At $I = 1$ the stable node and the saddle coalesce in a *saddle-node bifurcation of fixed points.*

Existence of a Closed Orbit

What happens when $I > 1$? There are no more fixed points available; something new has to happen. We claim that *all trajectories are attracted to a unique, stable limit cycle.*

The first step is to show that a periodic solution exists. The argument uses a clever idea introduced by Poincaré long ago. Watch carefully—this idea will come up frequently in our later work.

Consider the nullcline $y = \alpha^{-1}(I - \sin\phi)$ where $y' = 0$. The flow is downward above the nullcline and upward below it (Figure 8.5.1).

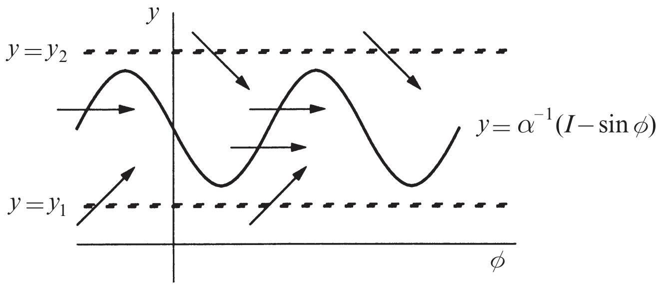

Figure 8.5.1

In particular, all trajectories eventually enter the strip $y_1 \le y \le y_2$ (Figure 8.5.1), and stay in there forever. (Here y_1 and y_2 are any fixed numbers such that $0 < y_1 < (I-1)/\alpha$ and $y_2 > (I+1)/\alpha$.) Inside the strip, the flow is always to the right, because $y > 0$ implies $\phi' > 0$.

Also, since $\phi = 0$ and $\phi = 2\pi$ are equivalent on the cylinder, we may as well confine our attention to the rectangular box $0 \le \phi \le 2\pi$, $y_1 \le y \le y_2$. This box contains all the information about the long-term behavior of the flow (Figure 8.5.2).

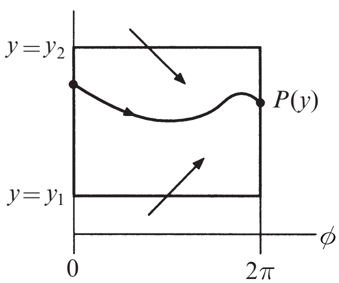

Figure 8.5.2

Now consider a trajectory that starts at a height y on the left side of the box, and follow it until it intersects the right side of the box at some new height $P(y)$, as shown in Figure 8.5.2. The mapping from y to $P(y)$ is called the ***Poincaré map***. It tells us how the height of a trajectory changes after one lap around the cylinder (Figure 8.5.3).

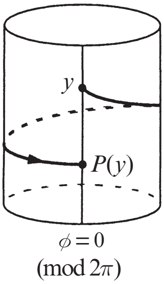

Figure 8.5.3

The Poincaré map is also called the ***first-return map***, because if a trajectory starts at a height y on the line $\phi = 0$ (mod 2π), then $P(y)$ is its height when it returns to that line for the first time.

Now comes the key point: we can't compute $P(y)$ explicitly, but *if we can show that there's a point y^* such that $P(y^*) = y^*$, then the corresponding trajectory will be a closed orbit* (because it returns to the same location on the cylinder after one lap).

To show that such a y^* must exist, we need to know what the graph of $P(y)$ looks like, at least roughly. Consider a trajectory that starts at $y = y_1$, $\phi = 0$. We claim that

$$P(y_1) > y_1.$$

This follows because the flow is strictly upward at first, and the trajectory can never return to the line $y = y_1$, since the flow is everywhere upward on that line (recall Figures 8.5.1 and 8.5.2). By the same kind of argument,

$$P(y_2) < y_2.$$

Furthermore, $P(y)$ is a *continuous* function. This follows from the theorem that solutions of differential equations depend continuously on initial conditions, if the vector field is smooth enough.

And finally, $P(y)$ is a *monotonic* function. (By drawing pictures, you can convince yourself that if $P(y)$ were not monotonic, two trajectories would cross—and that's forbidden.) Taken together, these results imply that $P(y)$ has the shape shown in Figure 8.5.4.

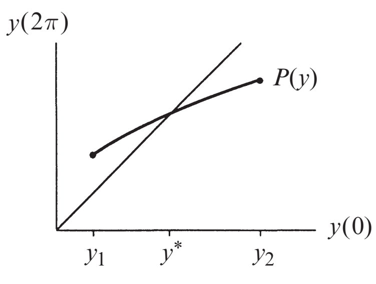

Figure 8.5.4

By the intermediate value theorem (or common sense), the graph of $P(y)$ must cross the 45° diagonal *somewhere*; that intersection is our desired y^*.

Uniqueness of the Limit Cycle

The argument above proves the *existence* of a closed orbit, and almost proves its uniqueness. But we haven't excluded the possibility that $P(y) \equiv y$ on some interval, in which case there would be a band of infinitely many closed orbits.

To nail down the uniqueness part of our claim, we recall from Section 6.7 that there are two topologically different kinds of periodic orbits on a cylinder: ***librations*** and ***rotations*** (Figure 8.5.5).

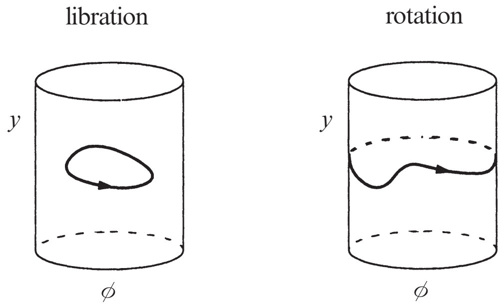

Figure 8.5.5

For $I > 1$, librations are impossible because any libration must encircle a fixed point, by index theory—but there are no fixed points when $I > 1$. Hence we only need to consider rotations.

Suppose there were two different rotations. The phase portrait on the cylinder would have to look like Figure 8.5.6.

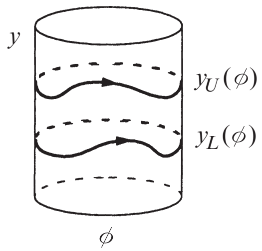

Figure 8.5.6

One of the rotations would have to lie *strictly above* the other because trajectories can't cross. Let $y_U(\phi)$ and $y_L(\phi)$ denote the "upper" and "lower" rotations, where $y_U(\phi) > y_L(\phi)$ for all ϕ.

The existence of two such rotations leads to a contradiction, as shown by the following energy argument. Let

$$E = \tfrac{1}{2} y^2 - \cos\phi. \tag{5}$$

After one circuit around any rotation $y(\phi)$, the change in energy ΔE must vanish. Hence

$$0 = \Delta E = \int_0^{2\pi} \frac{dE}{d\phi}\, d\phi. \tag{6}$$

But (5) implies

$$\frac{dE}{d\phi} = y\frac{dy}{d\phi} + \sin\phi \tag{7}$$

and

$$\frac{dy}{d\phi} = \frac{y'}{\phi'} = \frac{I - \sin\phi - \alpha y}{y}, \tag{8}$$

from (4). Substituting (8) into (7) gives $dE/d\phi = I - \alpha y$. Thus (6) implies

$$0 = \int_0^{2\pi} (I - \alpha y)\, d\phi$$

on any rotation $y(\phi)$. Equivalently, any rotation must satisfy

$$\int_0^{2\pi} y(\phi)\, d\phi = \frac{2\pi I}{\alpha}. \tag{9}$$

But since $y_U(\phi) > y_L(\phi)$,

$$\int_0^{2\pi} y_U(\phi)\, d\phi > \int_0^{2\pi} y_L(\phi)\, d\phi,$$

and so (9) can't hold for *both* rotations.

This contradiction proves that the rotation for $I > 1$ is unique, as claimed.

Homoclinic Bifurcation

Suppose we slowly decrease I, starting from some value $I > 1$. What happens to the rotating solution? Think about the pendulum: as the driving torque is reduced, the pendulum struggles more and more to make it over the top. At some critical value $I_c < 1$, the torque is insufficient to overcome gravity and damping, and the pendulum can no longer whirl. Then the rotation disappears and all solutions damp out to the rest state.

Our goal now is to visualize the corresponding bifurcation in phase space. In Exercise 8.5.2, you're asked to show (by numerical computation of the phase portrait) that if α is sufficiently small, the stable limit cycle is destroyed in a *homoclinic* *bifurcation* (Section 8.4). The following schematic drawings summarize the results you should get.

First suppose $I_c < I < 1$. The system is bistable: a sink coexists with a stable limit cycle (Figure 8.5.7).

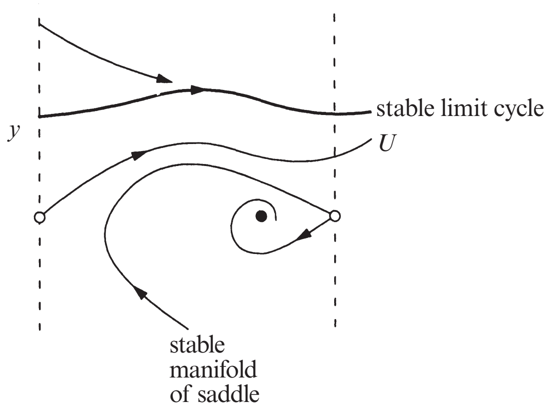

Figure 8.5.7

Keep your eye on the trajectory labeled U in Figure 8.5.7. It is a branch of the unstable manifold of the saddle. As $t \to \infty$, U asymptotically approaches the stable limit cycle.

As I decreases, the stable limit cycle moves down and squeezes U closer to the stable manifold of the saddle. When $I = I_c$, *the limit cycle merges with U* in a homoclinic bifurcation. Now U is a homoclinic orbit—it joins the saddle to itself (Figure 8.5.8).

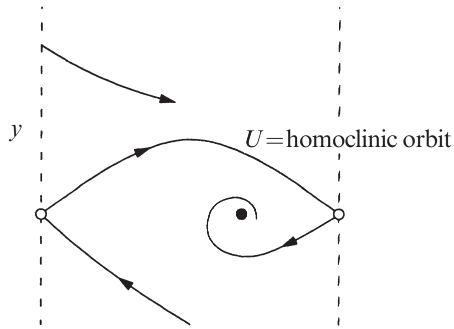

Figure 8.5.8

Finally, when $I < I_c$ the saddle connection breaks and U spirals into the sink (Figure 8.5.9).

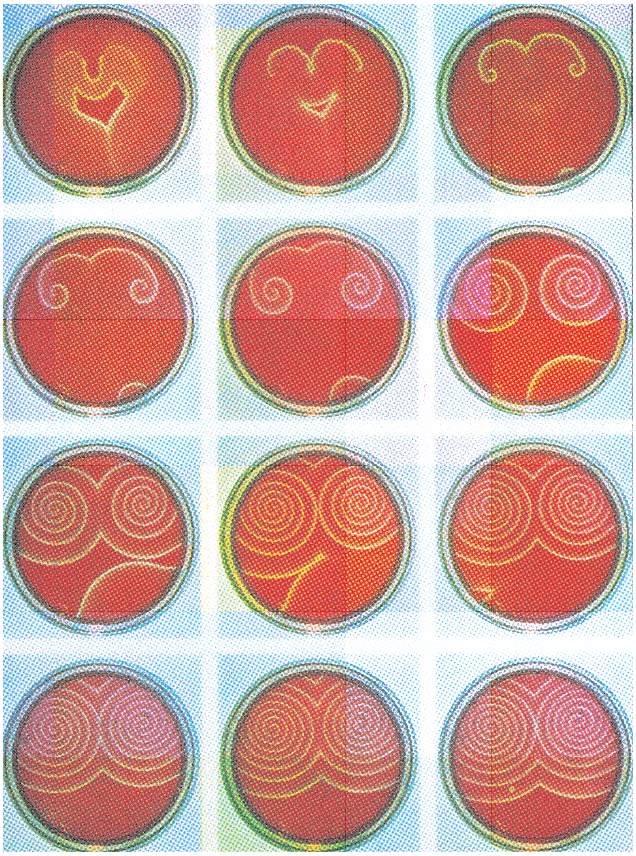

Plate 1: Spiral waves of chemical activity in a shallow dish of the Belousov–Zhabotinsky reaction (Section 8.3). These snapshots read from left to right and top to bottom. The complicated initial condition shown in the upper left was created by touching the liquid with a hot wire, thereby inducing an expanding circular wave of oxidation, and then disrupting this wave by gently rocking the dish. As time evolves, the blue waves propagate by diffusion through the motionless reddish-orange liquid. Whenever two waves collide, they annihilate each other, like grassfires rushing head on. Ultimately the system organizes itself into a pair of counterrotating spirals. Reproduced from Winfree (1974). Photographs by Fritz Goro.

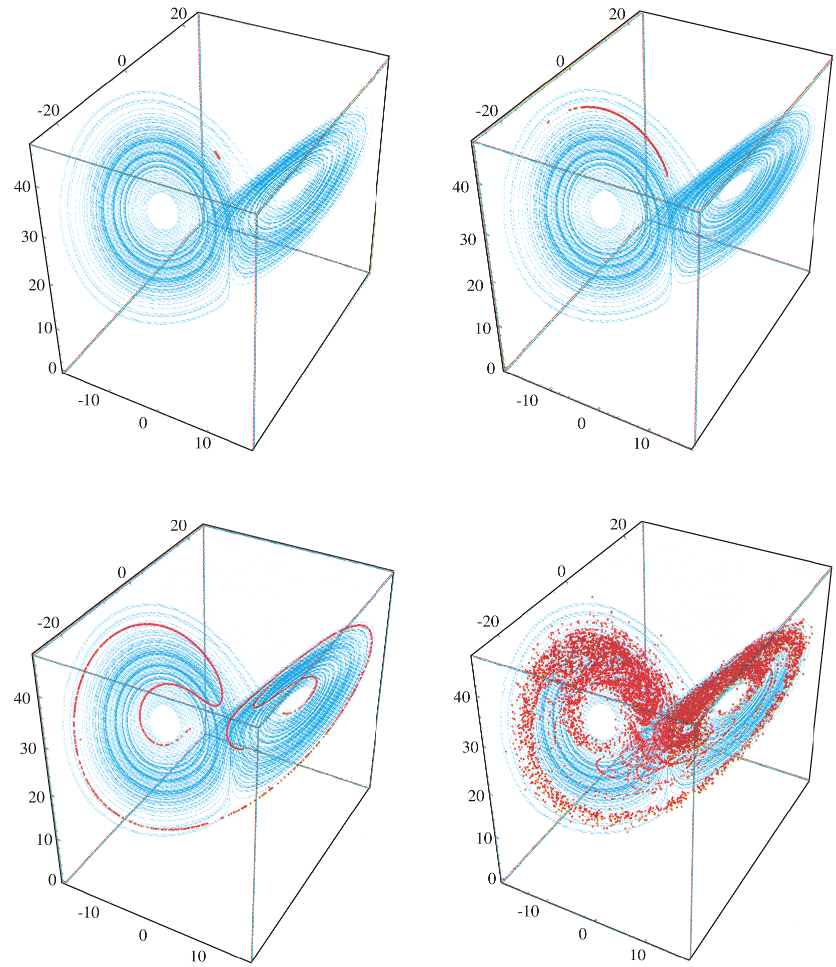

Plate 2: Divergence of nearby trajectories on the Lorenz attractor (Section 9.3). The Lorenz attractor is shown in blue. The red points show the evolution of a small blob of 10,000 nearby initial conditions, at times $t = 3$, 6, 9, and 15. As each point moves according to the Lorenz equations, the blob is stretched into a long thin filament, which then wraps around the attractor. Ultimately the points spread over much of the attractor, showing that the final state could be almost anywhere, even though the initial conditions were almost identical. This sensitive dependence on initial conditions is the signature of a chaotic system.

Plate inspired by a similar illustration in Crutchfield et al. (1986). Numerical integration and computer graphics by Thanos Siapas, using Equation (9.2.1) with parameters $\sigma = 10$, $b = 8/3$, $r = 28$.

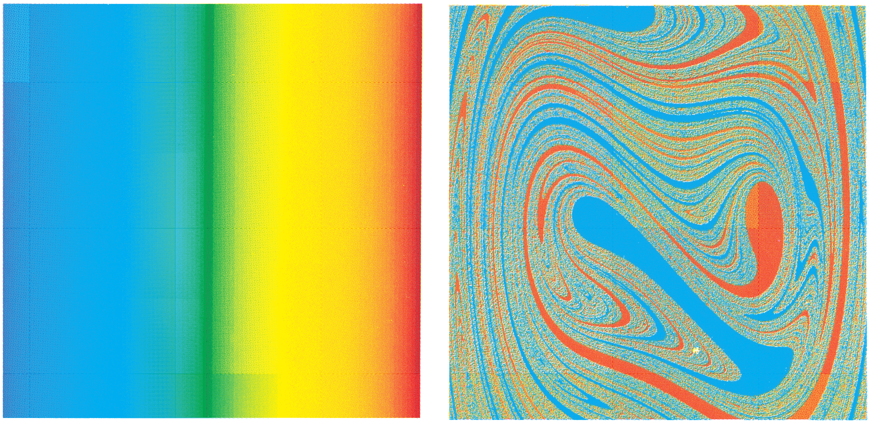

Plate 3: Fractal basin boundaries for the periodically forced double-well oscillator

$$x' = y, \qquad y' = x - x^3 - \delta y + F\cos\omega t,$$

with $\delta = 0.25$, $F = 0.25$, $\omega = 1$ (Section 12.5). For these parameter values, the system has two periodic attractors, corresponding to forced oscillations confined to the left or right well.

(a) Color map: The square region $-2.5 \le x, y \le 2.5$ is subdivided into 900×900 cells, and each cell is color-coded according to the x-position of its center point.

(b) Basins of attraction: Each cell is color-coded according to its fate after many drive cycles. Roughly speaking, if the trajectory ends up oscillating in the right well, the original cell is colored red; if it ends up in the left well, it is colored blue. More precisely, given an initial point (x_0, y_0) at the center of a cell, the state $(x(t), y(t))$ is computed at $t = 73 \times 2\pi/\omega$ (that is, after 73 drive cycles), and the original cell is color-coded by the value of $x(t)$. The basins have a complicated shape, and the boundary between them is fractal (Moon and Li 1985). Near the boundary, slight variations in initial conditions can lead to totally different outcomes.

Computations by Thanos Siapas on a Thinking Machines CM-5 parallel computer using a fifth-order Runge–Kutta–Fehlberg method.

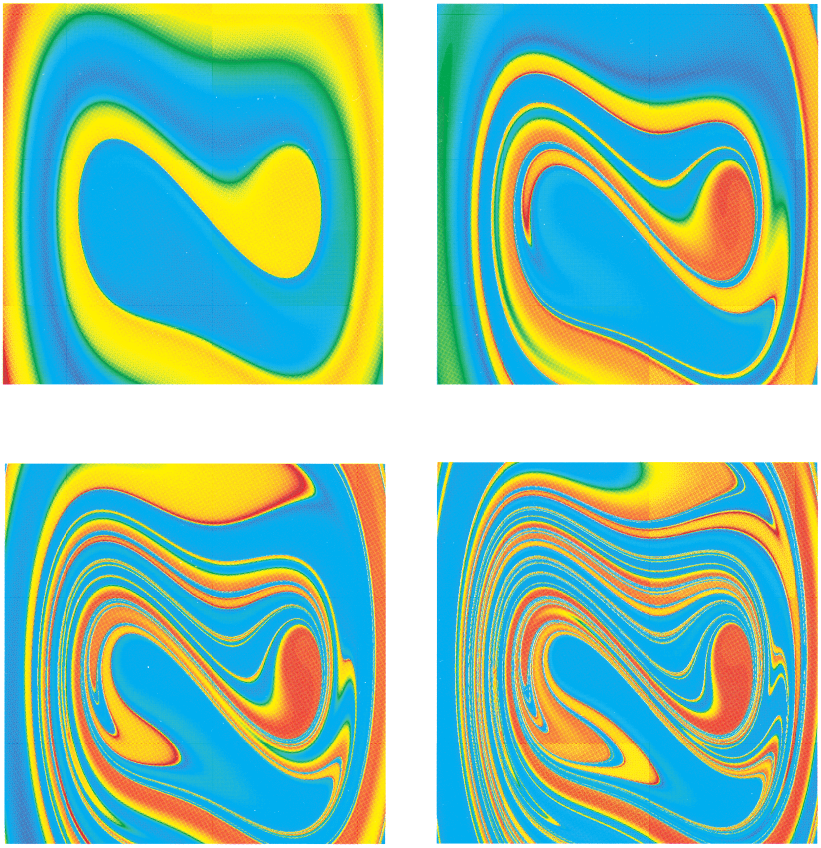

Plate 4: Maps of the short-term behavior of the periodically forced double-well oscillator. Equations, parameters, and color code as in Plate 3. However, instead of showing the system's asymptotic behavior, these plates show the color-coded value of $x(t)$ after only 1, 2, 3, and 4 drive cycles, respectively. The red and blue regions correspond to initial conditions that converge rapidly to one of the two attractors. A rainbow of colors is found near the basin boundary, because those initial conditions lead to trajectories that linger far from either attractor during the time shown.

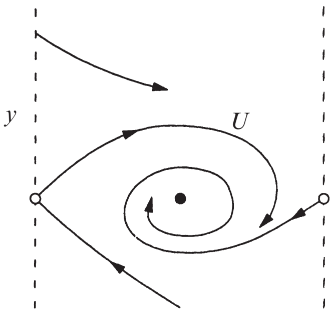

Figure 8.5.9

The scenario described here is valid only if the dimensionless damping α is sufficiently small. We know that something different has to happen for large α. After all, when α is infinite we are in the overdamped limit studied in Section 4.6. Our analysis there showed that the periodic solution is destroyed by an *infinite-period bifurcation* (a saddle and a node are born on the former limit cycle). So it's plausible that an infinite-period bifurcation should also occur if α is large but finite. These intuitive ideas are confirmed by numerical integration (Exercise 8.5.2).

Putting it all together, we arrive at the stability diagram shown in Figure 8.5.10. Three types of bifurcations occur: homoclinic and infinite-period bifurcations of periodic orbits, and a saddle-node bifurcation of fixed points.

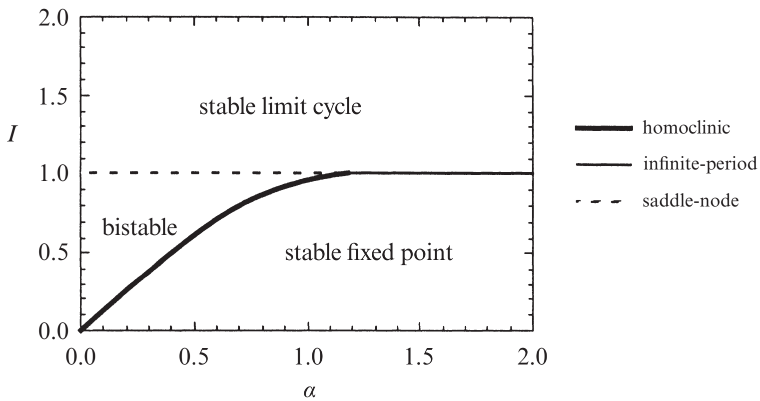

Figure 8.5.10

Our argument leading to Figure 8.5.10 has been heuristic. For rigorous proofs, see Levi et al. (1978). Also, Guckenheimer and Holmes (1983, p. 202) derive an analytical approximation for the homoclinic bifurcation curve for $\alpha << 1$, using an advanced technique known as Melnikov's method. They show that the bifurcation curve is tangent to the line $I = 4\alpha/\pi$ as $\alpha \to 0$. Even if α is not so small, this approximation works nicely, thanks to the straightness of the homoclinic bifurcation curve in Figure 8.5.10.

Hysteretic Current-Voltage Curve

Figure 8.5.10 explains why lightly damped Josephson junctions have hysteretic $I - V$ curves. Suppose α is small and I is initially below the homoclinic bifurcation (thick line in Figure 8.5.10). Then the junction will be operating at the stable fixed point, corresponding to the zero-voltage state. As I is increased, nothing changes until I exceeds 1. Then the stable fixed point disappears in a saddle-node bifurcation, and the junction jumps into a nonzero voltage state (the limit cycle).

If I is brought back down, the limit cycle persists below $I = 1$ but its frequency tends to zero continuously as I_c is approached. Specifically, the frequency tends to zero like $[\ln(I - I_c)]^{-1}$, just as expected from the scaling law discussed in Section 8.4. Now recall from Section 4.6 that the junction's dc-voltage is proportional to its oscillation frequency. Hence, the voltage also returns to zero continuously as $I \to I_c^+$ (Figure 8.5.11).

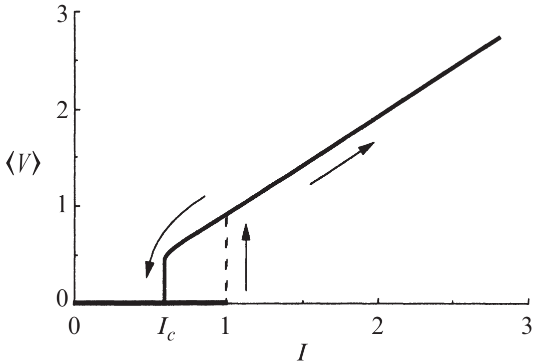

Figure 8.5.11

In practice, the voltage appears to jump discontinuously back to zero, but that is to be expected because $[\ln(I - I_c)]^{-1}$ has *infinite derivatives of all orders at* I_c ! (See Exercise 8.5.1.) The steepness of the curve makes it impossible to resolve the continuous return to zero. For instance, in experiments on pendula, Sullivan and Zimmerman (1971) measured the mechanical analog of the $I - V$ curve—namely, the curve relating the rotation rate to the applied torque. Their data show a jump back to zero rotation rate at the bifurcation.

8.6 Coupled Oscillators and Quasiperiodicity

Besides the plane and the cylinder, another important two-dimensional phase space is the ***torus***. It is the natural phase space for systems of the form

$$\dot{\theta}_1 = f_1(\theta_1, \theta_2)$$
$$\dot{\theta}_2 = f_2(\theta_1, \theta_2)$$

where f_1 and f_2 are periodic in both arguments.

For instance, a simple model of ***coupled oscillators*** is given by

$$\begin{aligned}\dot{\theta}_1 &= \omega_1 + K_1 \sin(\theta_2 - \theta_1)\\ \dot{\theta}_2 &= \omega_2 + K_2 \sin(\theta_1 - \theta_2),\end{aligned} \tag{1}$$

where θ_1, θ_2 are the *phases* of the oscillators, $\omega_1, \omega_2 > 0$ are their *natural frequencies*, and K_1, $K_2 \geq 0$ are *coupling constants*. Equation (1) has been used to model the interaction between human circadian rhythms and the sleep-wake cycle (Strogatz 1986, 1987).

An intuitive way to think about (1) is to imagine two friends jogging on a circular track. Here $\theta_1(t)$, $\theta_2(t)$ represent their positions on the track, and ω_1, ω_2 are proportional to their preferred running speeds. If they were uncoupled, then each would run at his or her preferred speed and the faster one would periodically overtake the slower one (as in Example 4.2.1). But these are *friends*—they want to run around *together*! So they need to compromise, with each adjusting his or her speed as necessary. If their preferred speeds are too different, phase-locking will be impossible and they may want to find new running partners.

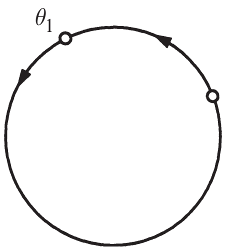

Figure 8.6.1

Here we consider (1) more abstractly, to illustrate some general features of flows on the torus and also to provide an example of a saddle-node bifurcation of cycles (Section 8.4). To visualize the flow, imagine two points running around a circle at instantaneous rates $\dot{\theta}_1, \dot{\theta}_2$ (Figure 8.6.1). Alternatively, we could imagine a *single* point tracing out a trajectory on a torus with coordinates θ_1, θ_2 (Figure 8.6.2). The coordinates are analogous to latitude and longitude.

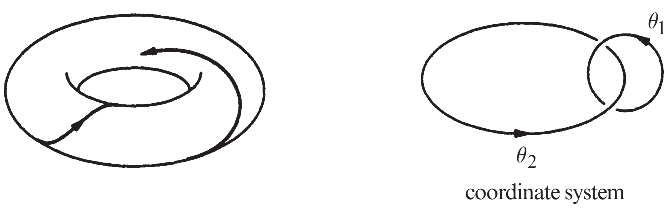

Figure 8.6.2

But since the curved surface of a torus makes it hard to draw phase portraits, we prefer to use an equivalent representation: a *square with periodic boundary conditions.* Then if a trajectory runs off an edge, it magically reappears on the opposite edge, as in some video games (Figure 8.6.3).

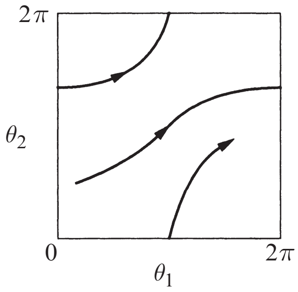

Figure 8.6.3

Uncoupled System

Even the seemingly trivial case of uncoupled oscillators ($K_1, K_2 = 0$) holds some surprises. Then (1) reduces to $\dot{\theta}_1 = \omega_1,\ \dot{\theta}_2 = \omega_2$. The corresponding trajectories on the square are straight lines with constant slope $d\theta_2/d\theta_1 = \omega_2/\omega_1$. There are two qualitatively different cases, depending on whether the slope is a rational or an irrational number.

If the slope is ***rational***, then $\omega_1/\omega_2 = p/q$ for some integers p, q with no common factors. In this case *all trajectories are closed orbits* on the torus, because θ_1 completes p revolutions in the same time that θ_2 completes q revolutions. For example, Figure 8.6.4 shows a trajectory on the square with $p = 3$, $q = 2$.

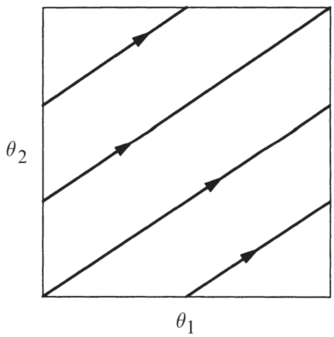

Figure 8.6.4

When plotted on the torus, the same trajectory gives . . . a ***trefoil knot***! Figure 8.6.5 shows a trefoil, alongside a top view of a torus with a trefoil wound around it.

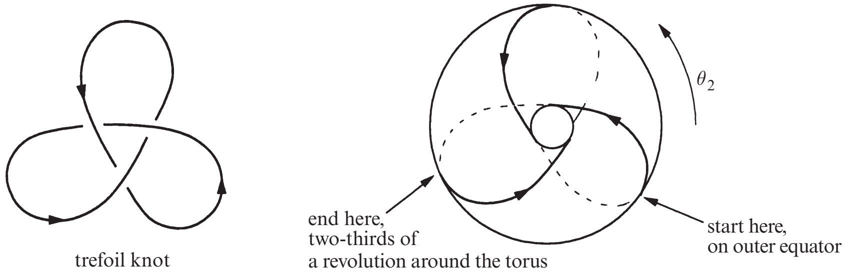

Figure 8.6.5

Do you see why this knot corresponds to $p = 3$, $q = 2$? Follow the knotted trajectory in Figure 8.6.5, and count the number of revolutions made by θ_2 during the time that θ_1 makes one revolution, where θ_1 is latitude and θ_2 is longitude. Starting on the outer equator, the trajectory moves onto the top surface, dives into the hole, travels along the bottom surface, and then reappears on the outer equator, *two-thirds* of the way around the torus. Thus θ_2 makes *two-thirds* of a revolution while θ_1 makes one revolution; hence $p = 3$, $q = 2$.

In fact the trajectories are always knotted if $p, q \geq 2$ have no common factors. The resulting curves are called *p:q torus knots.*

The second possibility is that the slope is ***irrational*** (Figure 8.6.6). Then the flow is said to be ***quasiperiodic***. Every trajectory winds around endlessly on the torus, never intersecting itself and yet never quite closing.

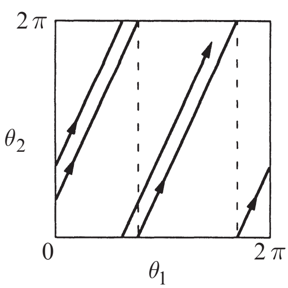

Figure 8.6.6

How can we be sure the trajectories never close? Any closed trajectory necessarily makes an integer number of revolutions in both θ_1 and θ_2; hence the slope would have to be rational, contrary to assumption.

Furthermore, when the slope is irrational, each trajectory is ***dense*** on the torus: in other words, each trajectory comes arbitrarily close to any given point on the torus. This is *not* to say that the trajectory passes *through* each point; it just comes arbitrarily close (Exercise 8.6.3).

Quasiperiodicity is significant because it is a new type of long-term behavior. Unlike the earlier entries (fixed point, closed orbit, homoclinic and heteroclinic orbits and cycles), quasiperiodicity occurs only on the torus.

Coupled System

Now consider (1) in the coupled case where $K_1, K_2 > 0$. The dynamics can be deciphered by looking at the *phase difference* $\phi = \theta_1 - \theta_2$. Then (1) yields

$$\dot{\phi} = \dot{\theta}_1 - \dot{\theta}_2$$
$$= \omega_1 - \omega_2 - (K_1 + K_2)\sin\phi, \quad (2)$$

which is just the nonuniform oscillator studied in Section 4.3. By drawing the standard picture (Figure 8.6.7), we see that there are two fixed points for (2) if $|\omega_1 - \omega_2| < K_1 + K_2$ and none if $|\omega_1 - \omega_2| > K_1 + K_2$. A saddle-node bifurcation occurs when $|\omega_1 - \omega_2| = K_1 + K_2$.

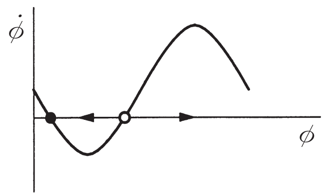

Figure 8.6.7

Suppose for now that there are two fixed points, defined implicitly by

$$\sin\phi^* = \frac{\omega_1 - \omega_2}{K_1 + K_2}.$$

As Figure 8.6.7 shows, all trajectories of (2) asymptotically approach the stable fixed point. Therefore, back on the torus, the trajectories of (1) approach a stable ***phase-locked*** solution in which the oscillators are separated by a constant phase difference ϕ^*. The phase-locked solution is *periodic*; in fact, both oscillators run at a constant frequency given by $\omega^* = \dot{\theta}_1 = \dot{\theta}_2 = \omega_2 + K_2 \sin\phi^*$. Substituting for $\sin\phi^*$ yields

$$\omega^* = \frac{K_1\omega_2 + K_2\omega_1}{K_1 + K_2}.$$

This is called the ***compromise frequency*** because it lies between the natural frequencies of the two oscillators (Figure 8.6.8).

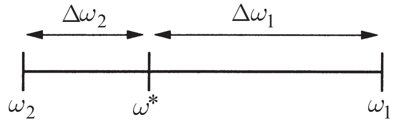

Figure 8.6.8

The compromise is not generally halfway; instead the frequencies are shifted by an amount proportional to the coupling strengths, as shown by the identity

$$\left|\frac{\Delta\omega_1}{\Delta\omega_2}\right| \equiv \left|\frac{\omega_1 - \omega^*}{\omega_2 - \omega^*}\right| = \left|\frac{K_1}{K_2}\right|.$$

Now we're ready to plot the phase portrait on the torus (Figure 8.6.9). The stable and unstable locked solutions appear as diagonal lines of slope 1, since $\dot{\theta}_1 = \dot{\theta}_2 = \omega^*$.

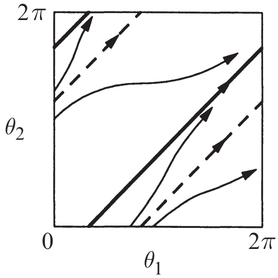

Figure 8.6.9

If we pull the natural frequencies apart, say by detuning one of the oscillators, then the locked solutions approach each other and coalesce when $|\omega_1 - \omega_2| = K_1 + K_2$. Thus the locked solution is destroyed in a *saddle-node bifurcation of cycles* (Section 8.4). After the bifurcation, the flow is like that in the uncoupled case studied earlier: we have either quasiperiodic or rational flow, depending on the parameters. The only difference is that now the trajectories on the square are curvy, not straight.

8.7 Poincaré Maps

In Section 8.5 we used a Poincaré map to prove the existence of a periodic orbit for the driven pendulum and Josephson junction. Now we discuss Poincaré maps more generally.

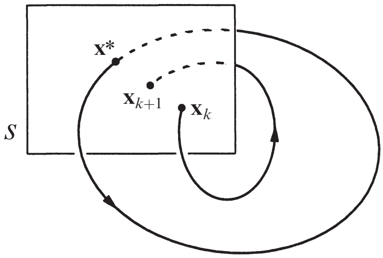

Figure 8.7.1

Poincaré maps are useful for studying swirling flows, such as the flow near a periodic orbit (or as we'll see later, the flow in some chaotic systems). Consider an n-dimensional system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$. Let S be an $n-1$ dimensional ***surface of section*** (Figure 8.7.1). S is required to be transverse to the flow, i.e., all trajectories starting on S flow through it, not parallel to it.

The ***Poincaré map*** P is a mapping from S to itself, obtained by following trajectories from one intersection with S to the next. If $\mathbf{x}_k \in S$ denotes the kth intersection, then the Poincaré map is defined by

$$\mathbf{x}_{k+1} = P(\mathbf{x}_k).$$

Suppose that $\mathbf{x}^*$ is *a **fixed point*** of P, i.e., $P(\mathbf{x}^*) = \mathbf{x}^*$. Then a trajectory starting at $\mathbf{x}^*$ returns to $\mathbf{x}^*$ after some time T, and is therefore *a closed orbit* for the original system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$. Moreover, by looking at the behavior of P near this fixed point, we can determine the stability of the closed orbit. Thus the Poincaré map converts problems about closed orbits (which are difficult) into problems about fixed points of a mapping (which are easier in principle, though not always in practice). The snag is that it's typically impossible to find a formula for P. For the sake of illustration, we begin with two examples for which P can be computed explicitly.

EXAMPLE 8.7.1:

Consider the vector field given in polar coordinates by $\dot{r} = r(1-r^2)$, $\dot{\theta} = 1$. Let S be the positive x-axis, and compute the Poincaré map. Show that the system has a unique periodic orbit and classify its stability.

Solution: Let r_0 be an initial condition on S. Since $\dot{\theta} = 1$, the first return to S occurs after a ***time of flight*** $t = 2\pi$. Then $r_1 = P(r_0)$, where r_1 satisfies

$$\int_{r_0}^{r_1} \frac{dr}{r(1-r^2)} = \int_0^{2\pi} dt = 2\pi.$$

Evaluation of the integral (Exercise 8.7.1) yields $r_1 = \left[1 + e^{-4\pi}(r_0^{-2} - 1)\right]^{-1/2}$. Hence $P(r) = \left[1 + e^{-4\pi}(r^{-2} - 1)\right]^{-1/2}$. The graph of P is plotted in Figure 8.7.2.

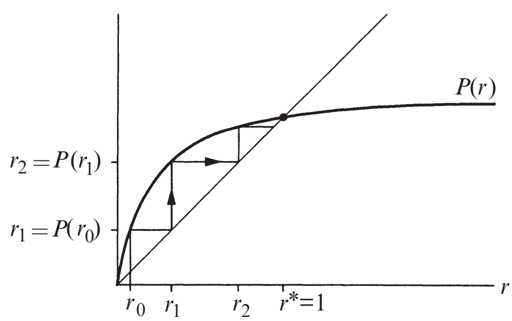

Figure 8.7.2

A fixed point occurs at $r^* = 1$ where the graph intersects the 45° line. The ***cobweb*** construction in Figure 8.7.2 enables us to iterate the map graphically. Given an input r_k, draw a vertical line until it intersects the graph of P; that height is the output r_{k+1}. To iterate, we make r_{k+1} the new input by drawing a horizontal line until it intersects the 45° diagonal line. Then repeat the process. Convince yourself that this construction works; we'll be using it often.

The cobweb shows that the fixed point $r^* = 1$ is stable and unique. No surprise, since we knew from Example 7.1.1 that this system has a stable limit cycle at $r = 1$. ■

EXAMPLE 8.7.2:

A sinusoidally forced *RC*-circuit can be written in dimensionless form as $\dot{x} + x = A \sin \omega t$, where $\omega > 0$. Using a Poincaré map, show that this system has a unique, globally stable limit cycle.

Solution: This is one of the few time-dependent systems we've discussed in this book. Such systems can always be made time-independent by adding a new variable. Here we introduce $\theta = \omega t$ and regard the system as a vector field on a cylinder: $\dot{\theta} = \omega,\ \dot{x} + x = A \sin \theta$. Any vertical line on the cylinder is an appropriate section S; we choose $S = \{(\theta, x)\colon \theta = 0 \bmod 2\pi\}$. Consider an initial condition on S given by $\theta(0) = 0$, $x(0) = x_0$. Then the time of flight between successive intersections is $t = 2\pi/\omega$. In physical terms, we strobe the system once per drive cycle and look at the consecutive values of x.

To compute P, we need to solve the differential equation. Its general solution is a sum of homogeneous and particular solutions: $x(t) = c_1 e^{-t} + c_2 \sin \omega t + c_3 \cos \omega t$. The constants c_2 and c_3 can be found explicitly, but the important point is that they depend on A and ω but *not* on the initial condition x_0; only c_1, depends on x_0. To make the dependence on x_0 explicit, observe that at $t = 0$, $x = x_0 = c_1 + c_3$. Thus

$$x(t) = (x_0 - c_3)e^{-t} + c_2 \sin \omega t + c_3 \cos \omega t.$$

Then P is defined by $x_1 = P(x_0) = x(2\pi/\omega)$. Substitution yields

$$\begin{aligned} P(x_0) &= x(2\pi/\omega) = (x_0 - c_3)e^{-2\pi/\omega} + c_3 \\ &= x_0 e^{-2\pi/\omega} + c_4 \end{aligned}$$

where $c_4 = c_3(1 - e^{-2\pi/\omega})$.

The graph of P is a straight line with a slope $e^{-2\pi/\omega} < 1$ as shown in Figure 8.7.3.

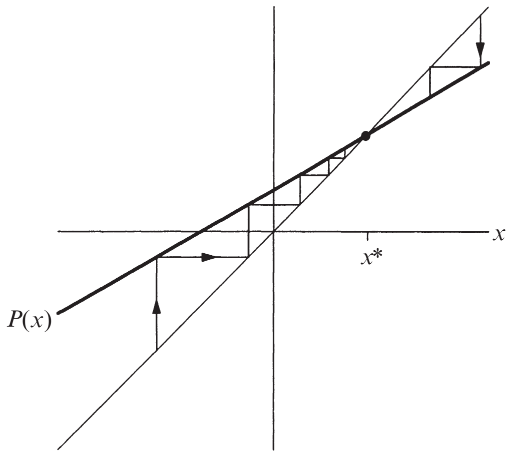

Figure 8.7.3

Since P has slope less than 1, it intersects the diagonal at a unique point. Furthermore, the cobweb shows that the deviation of x_k from the fixed point is reduced by a constant factor with each iteration. Hence the fixed point is unique and globally stable.

In physical terms, the circuit always settles into the same forced oscillation, regardless of the initial conditions. This is a familiar result from elementary physics, looked at in a new way. ■

Linear Stability of Periodic Orbits

Now consider the general case: Given a system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ with a closed orbit, how can we tell whether the orbit is stable is not? Equivalently, we ask whether the corresponding fixed point $\mathbf{x}^*$ of the Poincaré map is stable. Let $\mathbf{v}_0$ be an infinitesimal perturbation such that $\mathbf{x}^* + \mathbf{v}_0$ is in S. Then after the first return to S,

$$\begin{aligned} \mathbf{x}^* + \mathbf{v}_1 &= P(\mathbf{x}^* + \mathbf{v}_0) \\ &= P(\mathbf{x}^*) + \left[DP(\mathbf{x}^*)\right]\mathbf{v}_0 + O\left(\|\mathbf{v}_0\|^2\right) \end{aligned}$$

where $DP(\mathbf{x}^*)$ is an $(n-1) \times (n-1)$ matrix called the ***linearized Poincaré map*** at $\mathbf{x}^*$. Since $\mathbf{x}^* = P(\mathbf{x}^*)$, we get

$$\mathbf{v}_1 = [DP(\mathbf{x}^*)]\mathbf{v}_0$$

assuming that we can neglect the small $O\left(\|\mathbf{v}_0\|^2\right)$ terms.

The desired stability criterion is expressed in terms of the eigenvalues λ_j of $DP(\mathbf{x}^*)$: *The closed orbit is linearly stable if and only if* $|\lambda_j| < 1$ *for all* $j = 1, \ldots, n-1$.

To understand this criterion, consider the generic case where there are no repeated eigenvalues. Then there is a basis of eigenvectors $\{\mathbf{e}_j\}$ and so we can write $\mathbf{v}_0 = \sum_{j=1}^{n-1} v_j \mathbf{e}_j$ for some scalars v_j. Hence

$$\mathbf{v}_1 = \left(DP(\mathbf{x}^*)\right)\sum_{j=1}^{n-1} v_j \mathbf{e}_j = \sum_{j=1}^{n-1} v_j \lambda_j \mathbf{e}_j.$$

Iterating the linearized map k times gives

$$\mathbf{v}_k = \sum_{j=1}^{n-1} v_j (\lambda_j)^k \mathbf{e}_j.$$

Hence, if all $|\lambda_j| < 1$, then $\|\mathbf{v}_k\| \to 0$ geometrically fast. This proves that $\mathbf{x}^*$ is linearly stable. Conversely, if $|\lambda_j| > 1$ for some j, then perturbations along $\mathbf{e}_j$ grow, so $\mathbf{x}^*$ is unstable. A borderline case occurs when the largest eigenvalue has magnitude $|\lambda_m| = 1$; this occurs at bifurcations of periodic orbits, and then a nonlinear stability analysis is required.

The λ_j are called the ***characteristic*** or ***Floquet multipliers*** of the periodic orbit. (Strictly speaking, these are the *nontrivial* multipliers; there is always an additional trivial multiplier $\lambda \equiv 1$ corresponding to perturbations *along* the periodic orbit. We have ignored such perturbations since they just amount to time-translation.)

In general, the characteristic multipliers can only be found by numerical integration (see Exercise 8.7.10). The following examples are two of the rare exceptions.

EXAMPLE 8.7.3:

Find the characteristic multiplier for the limit cycle of Example 8.7.1.

Solution: We linearize about the fixed point $r^* = 1$ of the Poincaré map. Let $r = 1 + \eta$, where η is infinitesimal. Then $\dot{r} = \dot{\eta} = (1+\eta)(1-(1+\eta)^2)$. After neglecting $O(\eta^2)$ terms, we get $\dot{\eta} = -2\eta$. Thus $\eta(t) = \eta_0 e^{-2t}$. After a time of flight $t = 2\pi$, the new perturbation is $\eta_1 = e^{-4\pi}\eta_0$. Hence $e^{-4\pi}$ is the characteristic multiplier. Since $|e^{-4\pi}| < 1$, the limit cycle is linearly stable. ■

For this simple two-dimensional system, the linearized Poincaré map degenerates to a 1×1 matrix, i.e., a number. Exercise 8.7.1 asks you to show explicitly that $P'(r^*) = e^{-4\pi}$, as expected from the general theory above.

Our final example comes from an analysis of coupled Josephson junctions.

EXAMPLE 8.7.4:

The N-dimensional system

$$\dot{\phi}_i = \Omega + a\sin\phi_i + \tfrac{1}{N}\sum_{j=1}^{N}\sin\phi_j, \tag{1}$$

for $i = 1, \ldots, N$, describes the dynamics of a series array of overdamped Josephson junctions in parallel with a resistive load (Tsang et al. 1991). For technological reasons, there is great interest in the solution where all the junctions oscillate in phase. This *in-phase* solution is given by $\phi_1(t) = \phi_2(t) = \ldots = \phi_N(t) = \phi^*(t)$, where $\phi^*(t)$ denotes the common waveform. Find conditions under which the in-phase solution is periodic, and calculate the characteristic multipliers of this solution.

Solution: For the in-phase solution, all N equations reduce to

$$\frac{d\phi^*}{dt} = \Omega + (a+1)\sin\phi^*. \tag{2}$$

This has a periodic solution (on the circle) if and only if $|\Omega| > |a+1|$. To determine the stability of the in-phase solution, let $\phi_i(t) = \phi^*(t) + \eta_i(t)$, where the $\eta_i(t)$ are infinitesimal perturbations. Then substituting ϕ_i into (1) and dropping quadratic terms in η yields

$$\dot{\eta}_i = [a\cos\phi^*(t)]\eta_i + [\cos\phi^*(t)]\tfrac{1}{N}\sum_{j=1}^{N}\eta_j. \tag{3}$$

We don't have $\phi^*(t)$ explicitly, but that doesn't matter, thanks to two tricks. First, the linear system decouples if we change variables to

$$\mu = \tfrac{1}{N}\sum_{j=1}^{N}\eta_j,$$
$$\xi_i = \eta_{i+1} - \eta_i, \quad i = 1\ldots, N-1.$$

Then $\dot{\xi}_i = [a\cos\phi^*(t)]\xi_i$. Separation of variables yields

$$\frac{d\xi_i}{\xi_i} = [a\cos\phi^*(t)]dt = \frac{[a\cos\phi^*]d\phi^*}{\Omega + (a+1)\sin\phi^*},$$

where we've used (2) to eliminate dt. (That was the second trick.)

Now we compute the change in the perturbations after one circuit around the closed orbit ϕ^*:

$$\oint \frac{d\xi_i}{\xi_i} = \int_0^{2\pi} \frac{[a\cos\phi^*]d\phi^*}{\Omega+(a+1)\sin\phi^*}$$
$$\Rightarrow \ln\frac{\xi_i(T)}{\xi_i(0)} = \frac{a}{a+1} \ln\left[\Omega+(a+1)\sin\phi^*\right]_0^{2\pi} = 0.$$

Hence $\xi_i(T)=\xi_i(0)$. Similarly, we can show that $\mu(T)=\mu(0)$. Thus $\eta_i(T)=\eta_i(0)$ for all i; all perturbations are unchanged after one cycle! Therefore all the characteristic multipliers $\lambda_j = 1$. ■

This calculation shows that the in-phase state is (linearly) neutrally stable. That's discouraging technologically—one would like the array to lock into coherent oscillation, thereby greatly increasing the output power over that available from a single junction.

Since the calculation above is based on linearization, you might wonder whether the neglected nonlinear terms could stabilize the in-phase state. In fact they don't: a reversibility argument shows that the in-phase state is not attracting, even if the nonlinear terms are kept (Exercise 8.7.11).

EXERCISES FOR CHAPTER 8

8.1 Saddle-Node, Transcritical, and Pitchfork Bifurcations

8.1.1 For the following prototypical examples, plot the phase portraits as μ varies:

a) $\dot{x}=\mu x - x^2,\quad \dot{y}=-y$ (transcritical bifurcation)

b) $\dot{x}=\mu x + x^3,\quad \dot{y}=-y$ (subcritical pitchfork bifurcation)

For each of the following systems, find the eigenvalues at the stable fixed point as a function of μ, and show that one of the eigenvalues tends to zero as $\mu\to 0$.

8.1.2 $\dot{x}=\mu - x^2,\quad \dot{y}=-y$

8.1.3 $\dot{x}=\mu x - x^2,\quad \dot{y}=-y$

8.1.4 $\dot{x}=\mu x + x^3,\quad \dot{y}=-y$

8.1.5 True or false: at any zero-eigenvalue bifurcation in two dimensions, the nullclines always intersect tangentially. (Hint: Consider the geometrical meaning of the rows in the Jacobian matrix.)

8.1.6 Consider the system $\dot{x}=y-2x,\ \dot{y}=\mu+x^2-y$.

a) Sketch the nullclines.

b) Find and classify the bifurcations that occur as μ varies.

c) Sketch the phase portrait as a function of μ.

8.1.7 Find and classify all bifurcations for the system $\dot{x} = y - ax$, $\dot{y} = -by + x/(1+x)$.

8.1.8 (Bead on rotating hoop, revisited) In Section 3.5, we derived the following dimensionless equation for the motion of a bead on a rotating hoop:

$$\varepsilon \frac{d^2\phi}{d\tau^2} = -\frac{d\phi}{d\tau} - \sin\phi + \gamma \sin\phi \cos\phi.$$

Here $\varepsilon > 0$ is proportional to the mass of the bead, and $\gamma > 0$ is related to the spin rate of the hoop. Previously we restricted our attention to the overdamped limit $\varepsilon \to 0$.

a) Now allow any $\varepsilon > 0$. Find and classify all bifurcations that occur as ε and γ vary.
b) Plot the stability diagram in the positive quadrant of the ε,γ plane.

8.1.9 Plot the stability diagram for the system $\ddot{x} + b\dot{x} - kx + x^3 = 0$, where b and k can be positive, negative, or zero. Label the bifurcation curves in the (b, k) plane.

8.1.10 (Budworms vs. the forest) Ludwig et al. (1978) proposed a model for the effects of spruce budworm on the balsam fir forest. In Section 3.7, we considered the dynamics of the budworm population; now we turn to the dynamics of the forest. The condition of the forest is assumed to be characterized by $S(t)$, the average size of the trees, and $E(t)$, the "energy reserve" (a generalized measure of the forest's health). In the presence of a constant budworm population B, the forest dynamics are given by

$$\dot{S} = r_S S\left(1 - \frac{S}{K_S}\frac{K_E}{E}\right), \qquad \dot{E} = r_E E\left(1 - \frac{E}{K_E}\right) - P\frac{B}{S},$$

where r_S, r_E, K_S, K_E, $P > 0$ are parameters.

a) Interpret the terms in the model biologically.
b) Nondimensionalize the system.
c) Sketch the nullclines. Show that there are two fixed points if B is small, and none if B is large. What type of bifurcation occurs at the critical value of B?
d) Sketch the phase portrait for both large and small values of B.

8.1.11 In a study of isothermal autocatalytic reactions, Gray and Scott (1985) considered a hypothetical reaction whose kinetics are given in dimensionless form by

$$\dot{u} = a(1-u) - uv^2, \quad \dot{v} = uv^2 - (a+k)v,$$

where a, $k > 0$ are parameters. Show that saddle-node bifurcations occur at $k = -a \pm \frac{1}{2}\sqrt{a}$.

8.1.12 (Interacting bar magnets) Consider the system

$$\dot{\theta}_1 = K\sin(\theta_1 - \theta_2) - \sin\theta_1$$
$$\dot{\theta}_2 = K\sin(\theta_2 - \theta_1) - \sin\theta_2$$

where $K \geq 0$. For a rough physical interpretation, suppose that two bar magnets are confined to a plane, but are free to rotate about a common pin joint, as shown in Figure 1. Let θ_1, θ_2 denote the angular orientations of the north poles of the magnets. Then the term $K\sin(\theta_2 - \theta_1)$ represents a repulsive force that tries to keep the two north poles 180° apart. This repulsion is opposed by the $\sin\theta$ terms, which model external magnets that pull the north poles of both bar magnets to the east. If the inertia of the magnets is negligible compared to viscous damping, then the equations above are a decent approximation to the true dynamics.

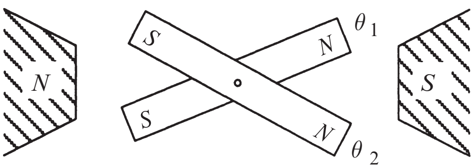

Figure 1

a) Find and classify all the fixed points of the system.
b) Show that a bifurcation occurs at $K = \frac{1}{2}$. What type of bifurcation is it? (Hint: Recall that $\sin(a-b) = \cos b \sin a - \sin b \cos a$.)
c) Show that the system is a "gradient" system, in the sense that $\dot{\theta}_i = -\partial V/\partial\theta_i$ for some potential function $V(\theta_1, \theta_2)$, to be determined.
d) Use part (c) to prove that the system has no periodic orbits.
e) Sketch the phase portrait for $0 < K < \frac{1}{2}$, and then for $K > \frac{1}{2}$.

8.1.13 (Laser model) In Exercise 3.3.1 we introduced the laser model

$$\dot{n} = GnN - kn$$
$$\dot{N} = -GnN - fN + p$$

where $N(t)$ is the number of excited atoms and $n(t)$ is the number of photons in the laser field. The parameter G is the gain coefficient for stimulated emission, k is the decay rate due to loss of photons by mirror transmission, scattering, etc., f is the decay rate for spontaneous emission, and p is the pump strength. All parameters are positive, except p, which can have either sign. For more information, see Milonni and Eberly (1988).

a) Nondimensionalize the system.
b) Find and classify all the fixed points.
c) Sketch all the qualitatively different phase portraits that occur as the dimensionless parameters are varied.
d) Plot the stability diagram for the system. What types of bifurcation occur?

8.1.14 (Binocular rivalry) Normally when you look at something, your left and right eyes see images that are very similar. (Try closing one eye, then the other; the resulting views look almost the same, except for the disparity caused by the spacing between your eyes.) But what would happen if two completely different images were shown to your left and right eyes simultaneously? What would you see? A combination of both images? Experiments like this have been performed for hundreds of years (Wade 1996), and the results are amazing: your brain typically perceives one image for a few seconds, then the other, then the first again, and so on. This switching phenomenon is known as *binocular rivalry.*

Mathematical models of binocular rivalry often posit that there are two neural populations corresponding to the brain's representations of the two competing images. These populations battle with each other for dominance—each tends to suppress the other. The following exercise, kindly suggested by Bard Ermentrout, involves the analysis of a minimal model for such neuronal competition.

Let x_1 and x_2 denote the averaged firing rates (essentially, the activity levels) of the two populations of neurons. Assume

$$\dot{x}_1 = -x_1 + F(I - bx_2), \quad \dot{x}_2 = -x_2 + F(I - bx_1),$$

where the gain function is given by $F(x) = 1/(1+e^{-x})$, I is the strength of the input stimulus (in this case, the stimuli are the images; note that each is assumed to be equally potent), and b is the strength of the mutual antagonism.

a) Sketch the phase plane for various values of I and b (both positive).
b) Show that the symmetric fixed point, $x_1^* = x_2^* = x^*$, is always a solution (in other words, it exists for all positive values of I and b), and show that it is unique.
c) Show that at a sufficiently large value of b, the symmetric solution loses stability at a pitchfork bifurcation. Which type of pitchfork bifurcation is it?

For a refinement of this model that allows for rhythmic switching between the two perceived images, see Exercise 8.2.17. For more elaborate models and a comparative study of their bifurcation structure, see Shpiro et al. (2007).

8.1.15 (The power of true believers) Sometimes a small band of unwavering advocates can win an entire population over to their point of view, as in the case of the civil rights or women's suffrage movements in the United States. Consider the following stylized model of such situations, which was studied by Marvel et al. (2012) and inspired by the earlier work of Xie et al. (2011).

The population is divided into four non-overlapping groups. An initially small group of true believers holds opinion A (for example, that women deserve the right to vote) and they are committed to this belief. Nothing that anyone says or does can change their minds. Another group of people currently agrees with them, but they are uncommitted to A. If someone argues for the opposing position, B, an uncommitted A-believer instantly becomes an AB, meaning someone who sees the merit in both positions. Likewise, B-believers instantly turn into members of the AB subpopulation when confronted with an argument for A. The people in the AB group, being fence-sitters, don't try to persuade anyone of anything. And they can be pushed to either side with the slightest nudge; when confronted with an argument for A—or for B—they join that camp. At each time step, we select two people at random and have one of them act as an advocate and the other as a listener. Assuming the members of the four groups mix with each other at random, the governing equations for the dynamics are

$$\dot{n}_A = (p + n_A)n_{AB} - n_A n_B$$
$$\dot{n}_B = n_B n_{AB} - (p + n_A)n_{AB}$$

where $n_{AB} = 1 - (p + n_A) - n_B$. Here the parameter p denotes the unchanging fraction of true believers in the population. The time-dependent variables n_A, n_B, and n_{AB} are the current fractions in the A, B, and AB subpopulations.

a) Interpret and justify the form of the various terms in the governing equations.
b) Assume that initially everyone believes in B, except for the true believers in A. Thus, $n_B(0) = 1 - p$, and $n_A(0) = n_{AB}(0) = 0$. Numerically integrate the system until it reaches equilibrium. Show that the final state changes discontinuously as a function of p. Specifically, show there is a critical fraction of true believers (call it p_c) such that for $p < p_c$, most people still accept B, whereas for $p > p_c$, everyone comes around to A.
c) Show analytically that $p_c = 1 - \sqrt{3}/2 \approx 0.134$. Thus, in this model, only about 13% of the population needs to be unwavering advocates to get everyone else to agree with them eventually.
d) What type of bifurcation occurs at p_c?

8.2 Hopf Bifurcations

8.2.1 Consider the biased van der Pol oscillator $\ddot{x} + \mu(x^2 - 1)\dot{x} + x = a$. Find the curves in (μ, a) space at which Hopf bifurcations occur.

The next three exercises deal with the system $\dot{x} = -y + \mu x + xy^2$, $\dot{y} = x + \mu y - x^2$.

8.2.2 By calculating the linearization at the origin, show that the system $\dot{x} = -y + \mu x + xy^2$, $\dot{y} = x + \mu y - x^2$ has pure imaginary eigenvalues when $\mu = 0$.

8.2.3 (Computer work) By plotting phase portraits on the computer, show that the system $\dot{x} = -y + \mu x + xy^2$, $\dot{y} = x + \mu y - x^2$ undergoes a Hopf bifurcation at $\mu = 0$. Is it subcritical, supercritical, or degenerate?

8.2.4 (A heuristic analysis) The system $\dot{x} = -y + \mu x + xy^2$, $\dot{y} = x + \mu y - x^2$ can be analyzed in a rough, intuitive way as follows.

a) Rewrite the system in polar coordinates.
b) Show that if $r << 1$, then $\dot{\theta} \approx 1$ and $\dot{r} \approx \mu r + \frac{1}{8}r^3 + \ldots$, where the terms omitted are oscillatory and have essentially zero time-average around one cycle.
c) The formulas in part (b) suggest the presence of an unstable limit cycle of radius $r \approx \sqrt{-8\mu}$ for $\mu < 0$. Confirm that prediction numerically. (Since we assumed that $r << 1$, the prediction is expected to hold only if $|\mu| << 1$.)

The reasoning above is shaky. See Drazin (1992, pp. 188-190) for a proper analysis via the Poincaré-Lindstedt method.

For each of the following systems, a Hopf bifurcation occurs at the origin when $\mu = 0$. Using a computer, plot the phase portrait and determine whether the bifurcation is subcritical or supercritical.

8.2.5 $\dot{x} = y + \mu x$, $\dot{y} = -x + \mu y - x^2 y$

8.2.6 $\dot{x} = \mu x + y - x^3$, $\dot{y} = -x + \mu y - 2y^3$

8.2.7 $\dot{x} = \mu x + y - x^2$, $\dot{y} = -x + \mu y - 2x^2$

8.2.8 (Predator-prey model) Odell (1980) considered the system

$$\dot{x} = x\left[x(1-x) - y\right], \quad \dot{y} = y(x-a),$$

where $x \geq 0$ is the dimensionless population of the prey, $y \geq 0$ is the dimensionless population of the predator, and $a \geq 0$ is a control parameter.

a) Sketch the nullclines in the first quadrant $x, y \geq 0$.
b) Show that the fixed points are (0, 0), (1, 0), and $(a, a - a^2)$, and classify them.
c) Sketch the phase portrait for $a > 1$, and show that the predators go extinct.
d) Show that a Hopf bifurcation occurs at $a_c = \frac{1}{2}$. Is it subcritical or supercritical?
e) Estimate the frequency of limit cycle oscillations for a near the bifurcation.
f) Sketch all the topologically different phase portraits for $0 < a < 1$.

The article by Odell (1980) is worth looking up. It is an outstanding pedagogical introduction to the Hopf bifurcation and phase plane analysis in general.

8.2.9 Consider the predator-prey model

$$\dot{x} = x\left(b - x - \frac{y}{1+x}\right), \quad \dot{y} = y\left(\frac{x}{1+x} - ay\right),$$

where $x, y \geq 0$ are the populations and $a, b > 0$ are parameters.

a) Sketch the nullclines and discuss the bifurcations that occur as b varies.
b) Show that a positive fixed point $x^* > 0, y^* > 0$ exists for all $a, b > 0$. (Don't try to find the fixed point explicitly; use a graphical argument instead.)
c) Show that a Hopf bifurcation occurs at the positive fixed point if

$$a = a_c = \frac{4(b-2)}{b^2(b+2)}$$

and $b > 2$. (Hint: A necessary condition for a Hopf bifurcation to occur is $\tau = 0$, where τ is the trace of the Jacobian matrix at the fixed point. Show that $\tau = 0$ if and only if $2x^* = b - 2$. Then use the fixed point conditions to express a_c in terms of x^*. Finally, substitute $x^* = (b-2)/2$ into the expression for a_c and you're done.)

d) Using a computer, check the validity of the expression in (c) and determine whether the bifurcation is subcritical or supercritical. Plot typical phase portraits above and below the Hopf bifurcation.

8.2.10 (Bacterial respiration) Fairén and Velarde (1979) considered a model for respiration in a bacterial culture. The equations are

$$\dot{x} = B - x - \frac{xy}{1+qx^2}, \qquad \dot{y} = A - \frac{xy}{1+qx^2}$$

where x and y are the levels of nutrient and oxygen, respectively, and A, B, $q > 0$ are parameters. Investigate the dynamics of this model. As a start, find all the fixed points and classify them. Then consider the nullclines and try to construct a trapping region. Can you find conditions on A, B, q under which the system has a stable limit cycle? Use numerical integration, the Poincaré-Bendixson theorem, results about Hopf bifurcations, or whatever else seems useful. (This question is deliberately open-ended and could serve as a class project; see how far you can go.)

8.2.11 (Degenerate bifurcation, not Hopf) Consider the damped Duffing oscillator $\ddot{x} + \mu\dot{x} + x - x^3 = 0$.

a) Show that the origin changes from a stable to an unstable spiral as μ decreases though zero.
b) Plot the phase portraits for $\mu > 0$, $\mu = 0$, and $\mu < 0$, and show that the bifurcation at $\mu = 0$ is a degenerate version of the Hopf bifurcation.

8.2.12 (Analytical criterion to decide if a Hopf bifurcation is subcritical or supercritical) Any system at a Hopf bifurcation can be put into the following form by suitable changes of variables:

$$\dot{x} = -\omega y + f(x,y), \qquad \dot{y} = \omega x + g(x,y),$$

where f and g contain only higher-order nonlinear terms that vanish at the origin. As shown by Guckenheimer and Holmes (1983, pp. 152-156), one can decide whether the bifurcation is subcritical or supercritical by calculating the sign of the following quantity:

$$16a = f_{xxx} + f_{xyy} + g_{xxy} + g_{yyy} + \frac{1}{\omega}\left[f_{xy}(f_{xx} + f_{yy}) - g_{xy}(g_{xx} + g_{yy}) - f_{xx}g_{xx} + f_{yy}g_{yy}\right]$$

where the subscripts denote partial derivatives evaluated at (0,0). The criterion is: If $a < 0$, the bifurcation is supercritical; if $a > 0$, the bifurcation is subcritical.

a) Calculate a for the system $\dot{x} = -y + xy^2, \quad \dot{y} = x - x^2$.

b) Use part (a) to decide which type of Hopf bifurcation occurs for $\dot{x} = -y + \mu x + xy^2, \quad \dot{y} = x + \mu y - x^2$ at $\mu = 0$. (Compare the results of Exercises 8.2.2-8.2.4.)

(You might be wondering what a measures. Roughly speaking, a is the coefficient of the cubic term in the equation $\dot{r} = ar^3$ governing the radial dynamics at the bifurcation. Here r is a slightly transformed version of the usual polar coordinate. For details, see Guckenheimer and Holmes (1983) or Grimshaw (1990).)

For each of the following systems, a Hopf bifurcation occurs at the origin when $\mu = 0$. Use the analytical criterion of Exercise 8.2.12 to decide if the bifurcation is sub- or supercritical. Confirm your conclusions on the computer.

8.2.13 $\dot{x} = y + \mu x, \quad \dot{y} = -x + \mu y - x^2 y$

8.2.14 $\dot{x} = \mu x + y - x^3, \quad \dot{y} = -x + \mu y + 2y^3$

8.2.15 $\dot{x} = \mu x + y - x^2, \quad \dot{y} = -x + \mu y + 2x^2$

8.2.16 In Example 8.2.1, we argued that the system $\dot{x} = \mu x - y + xy^2$, $\dot{y} = x + \mu y + y^3$ undergoes a subcritical Hopf bifurcation at $\mu = 0$. Use the analytical criterion to confirm that the bifurcation is subcritical.

8.2.17 (Binocular rivalry revisited) Exercise 8.1.14 introduced a minimal model of binocular rivalry, a remarkable visual phenomenon in which, when two different images are presented to the left and right eyes, the brain perceives only one image at a time. First one image wins, then the other, then the first again, and so on, with the winning image alternating every few seconds. The model studied in Exercise 8.1.14 could account for the complete suppression of one image by the other, but not for the rhythmic alternation between them. Now we extend that model to allow for the observed oscillations. (Many thanks to Bard Ermentrout for creating and sharing this exercise and the earlier Exercise 8.1.14.)

Let x_1 and x_2 denote the activity levels of the two populations of neurons coding for the two images, as in Exercise 8.1.14, but now assume the neurons get tired after winning a while, as adaptation builds up. The governing equations then become

$$\begin{aligned}
\dot{x}_1 &= -x_1 + F(I - bx_2 - gy_1) \\
\dot{y}_1 &= (-y_1 + x_1)/T \\
\dot{x}_2 &= -x_2 + F(I - bx_1 - gy_2) \\
\dot{y}_2 &= (-y_2 + x_2)/T
\end{aligned}$$

where the y-variables represent adaptation building up on a time scale T and provoking tiredness with strength g in their associated neuronal population. As in Exercise 8.1.14, the gain function is given by $F(x) = 1/(1+e^{-x})$, I is the strength of the input stimuli (the images), and b is the strength of the mutual antagonism between the neuronal populations. This is now a four-dimensional system, but its key stability properties can be inferred from two-dimensional calculations as follows.

a) Show that $x_1^* = y_1^* = x_2^* = y_2^* = u$ is a fixed point for all choices of parameters and that u is uniquely defined.

b) Show that the stability matrix (the Jacobian) for the linearization about this fixed point has the form

$$\begin{pmatrix} -c_1 & -c_2 & -c_3 & 0 \\ d_1 & -d_1 & 0 & 0 \\ -c_3 & 0 & -c_1 & -c_2 \\ 0 & 0 & d_1 & -d_1 \end{pmatrix}.$$

We can write this in block-matrix form as

$$\begin{pmatrix} A & B \\ B & A \end{pmatrix}$$

where A and B are 2×2 matrices. Show that the four eigenvalues of the 4×4 block matrix are given by the eigenvalues of $A-B$ and $A+B$.

c) Show that the eigenvalues of $A+B$ are all negative, by considering the trace and determinant of this matrix.

d) Show that depending on the sizes of g and T, the matrix $A-B$ can have either a negative determinant (leading to a pitchfork bifurcation of u) or a positive trace (resulting in a Hopf bifurcation).

e) Using a computer, show that the Hopf bifurcation can be supercritical; the resulting stable limit cycle mimics the oscillations we're trying to explain. (Hint: When looking for trajectories that approach a stable limit cycle, be sure to use initial conditions that are different for populations 1 and 2. In other words, break the symmetry by starting with $x_1 \neq x_2$ and $y_1 \neq y_2$.)

8.3 Oscillating Chemical Reactions

8.3.1 (Brusselator) The Brusselator is a simple model of a hypothetical chemical oscillator, named after the home of the scientists who proposed it. (This is a common joke played by the chemical oscillator community; there is also the "Oregonator," "Palo Altonator," etc.) In dimensionless form, its kinetics are

$$\dot{x} = 1-(b+1)x+ax^2y$$
$$\dot{y} = bx-ax^2y$$

where $a, b > 0$ are parameters and $x, y \geq 0$ are dimensionless concentrations.

a) Find all the fixed points, and use the Jacobian to classify them.
b) Sketch the nullclines, and thereby construct a trapping region for the flow.
c) Show that a Hopf bifurcation occurs at some parameter value $b = b_c$, where b_c is to be determined.
d) Does the limit cycle exist for $b > b_c$ or $b < b_c$? Explain, using the Poincaré-Bendixson theorem.
e) Find the approximate period of the limit cycle for $b \approx b_c$.

8.3.2 Schnackenberg (1979) considered the following hypothetical model of a chemical oscillator:

$$X \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} A, \qquad B \xrightarrow{k_2} Y, \qquad 2X+Y \xrightarrow{k_3} 3X.$$

After using the Law of Mass Action and nondimensionalizing, Schnackenberg reduced the system to

$$\dot{x} = a-x+x^2y$$
$$\dot{y} = b-x^2y$$

where $a, b > 0$ are parameters and $x, y > 0$ are dimensionless concentrations.

a) Show that all trajectories eventually enter a certain trapping region, to be determined. Make the trapping region as small as possible. (Hint: Examine the ratio $\dot{y}/\dot{x}$ for large x.)
b) Show that the system has a unique fixed point, and classify it.
c) Show that the system undergoes a Hopf bifurcation when $b - a = (a+b)^3$.
d) Is the Hopf bifurcation subcritical or supercritical? Use a computer to decide.
e) Plot the stability diagram in a, b space. (Hint: It is a bit confusing to plot the curve $b - a = (a+b)^3$, since this requires analyzing a cubic. As in Section 3.7, the *parametric form* of the bifurcation curve comes to the rescue. Show that the bifurcation curve can be expressed as

$$a = \tfrac{1}{2}x^*\left(1-(x^*)^2\right), \qquad b = \tfrac{1}{2}x^*\left(1+(x^*)^2\right)$$

where $x^* > 0$ is the x-coordinate of the fixed point. Then plot the bifurcation curve from these parametric equations. This trick is discussed in Murray (2002).)

8.3.3 (Relaxation limit of a chemical oscillator) Analyze the model for the chlorine dioxide–iodine–malonic acid oscillator, (8.3.4), (8.3.5), in the limit $b << 1$. Sketch the limit cycle in the phase plane and estimate its period.

8.4 Global Bifurcations of Cycles

8.4.1 Consider the system $\dot{r} = r(1-r^2)$, $\dot{\theta} = \mu - \sin\theta$ for μ slightly greater than 1. Let $x = r\cos\theta$ and $y = r\sin\theta$. Sketch the waveforms of $x(t)$ and $y(t)$. (These are typical of what one might see experimentally for a system on the verge of an infinite-period bifurcation.)

8.4.2 Discuss the bifurcations of the system $\dot{r} = r(\mu - \sin r)$, $\dot{\theta} = 1$ as μ varies.

8.4.3 (Homoclinic bifurcation) Using numerical integration, find the value of μ at which the system $\dot{x} = \mu x + y - x^2$, $\dot{y} = -x + \mu y + 2x^2$ undergoes a homoclinic bifurcation. Sketch the phase portrait just above and below the bifurcation.

8.4.4 (Second-order phase-locked loop) Using a computer, explore the phase portrait of $\ddot{\theta} + (1 - \mu\cos\theta)\dot{\theta} + \sin\theta = 0$ for $\mu \geq 0$. For some values of μ, you should find that the system has a stable limit cycle. Classify the bifurcations that create and destroy the cycle as μ increases from 0.

Exercises 8.4.5-8.4.11 deal with the ***forced Duffing oscillator*** in the limit where the forcing, detuning, damping, and nonlinearity are all weak:

$$\ddot{x} + x + \varepsilon(bx^3 + k\dot{x} - ax - F\cos t) = 0,$$

where $0 < \varepsilon << 1$, $b > 0$ is the nonlinearity, $k > 0$ is the damping, a is the detuning, and $F > 0$ is the forcing strength. This system is a small perturbation of a harmonic oscillator, and can therefore be handled with the methods of Section 7.6. We have postponed the problem until now because saddle-node bifurcations of cycles arise in its analysis.

8.4.5 (Averaged equations) Show that the averaged equations (7.6.53) for the system are

$$r' = -\tfrac{1}{2}(kr + F\sin\phi), \qquad \phi' = -\tfrac{1}{8}(4a - 3br^2 + \tfrac{4F}{r}\cos\phi),$$

where $x = r\cos(t+\phi)$, $\dot{x} = -r\sin(t+\phi)$, and prime denotes differentiation with respect to slow time $T = \varepsilon t$, as usual. (If you skipped Section 7.6, accept these equations on faith.)

8.4.6 (Correspondence between averaged and original systems) Show that fixed points for the averaged system correspond to phase-locked periodic solutions for the original forced oscillator. Show further that saddle-node bifurcations of fixed points for the averaged system correspond to saddle-node bifurcations of cycles for the oscillator.

8.4.7 (No periodic solutions for averaged system) Regard (r,ϕ) as polar coordinates in the phase plane. Show that the averaged system has no closed orbits. (Hint: Use Dulac's criterion with $g(r,\phi) \equiv 1$. Let $\mathbf{x}' = (r', r\phi')$. Compute $\nabla \cdot \mathbf{x}' = \frac{1}{r}\frac{\partial}{\partial r}(rr') + \frac{1}{r}\frac{\partial}{\partial \phi}(r\phi')$ and show that it has one sign.)

8.4.8 (No sources for averaged system) The result of the previous exercise shows that we only need to study the fixed points of the averaged system to determine its long-term behavior. Explain why the divergence calculation above also implies that the fixed points cannot be sources; only sinks and saddles are possible.

8.4.9 (Resonance curves and cusp catastrophe) In this exercise you are asked to determine how the equilibrium amplitude of the driven oscillations depends on the other parameters.

a) Show that the fixed points satisfy $r^2\left[k^2 + (\frac{3}{4}br^2 - a)^2\right] = F^2$.
b) From now on, assume that k and F are fixed. Graph r vs. a for the linear oscillator ($b = 0$). This is the familiar resonance curve.
c) Graph r vs. a for the nonlinear oscillator ($b \neq 0$). Show that the curve is single-valued for small nonlinearity, say $b < b_c$, but triple-valued for large nonlinearity ($b > b_c$), and find an explicit formula for b_c. (Thus we obtain the intriguing conclusion that the driven oscillator can have three limit cycles for some values of a and b !)
d) Show that if r is plotted as a surface above the (a, b) plane, the result is a cusp catastrophe surface (recall Section 3.6).

8.4.10 Now for the hard part: analyze the bifurcations of the averaged system.

a) Plot the nullclines $r' = 0$ and $\phi' = 0$ in the phase plane, and study how their intersections change as the detuning a is increased from negative values to large positive values.
b) Assuming that $b > b_c$, show that as a increases, the number of *stable* fixed points changes from one to two and then back to one again.

8.4.11 (Numerical exploration) Fix the parameters $k = 1$, $b = \frac{4}{3}$, $F = 2$.

a) Using numerical integration, plot the phase portrait for the averaged system with a increasing from negative to positive values.
b) Show that for $a = 2.8$, there are two stable fixed points.
c) Go back to the original forced Duffing equation. Numerically integrate it and plot $x(t)$ as a increases slowly from $a = -1$ to $a = 5$, and then decreases slowly back to $a = -1$. You should see a dramatic hysteresis effect with the limit cycle oscillation suddenly jumping up in amplitude at one value of a, and then back down at another.

8.4.12 (Scaling near a homoclinic bifurcation) To find how the period of a closed orbit scales as a homoclinic bifurcation is approached, we estimate the time it takes for a trajectory to pass by a saddle point (this time is much longer than all others in the problem). Suppose the system is given locally by $\dot{x} \approx \lambda_u x,\ \dot{y} \approx -\lambda_s y$. Let a trajectory pass through the point $(\mu, 1)$, where $\mu << 1$ is the distance from the stable manifold. How long does it take until the trajectory has escaped from the saddle, say out to $x(t) \approx 1$? (See Gaspard (1990) for a detailed discussion.)

8.5 Hysteresis in the Driven Pendulum and Josephson Junction

8.5.1 Show that $[\ln(I - I_c)]^{-1}$ has infinite derivatives of all orders at I_c. (Hint: Consider $f(I) = (\ln I)^{-1}$ and try to derive a formula for $f^{(n+1)}(I)$ in terms of $f^{(n)}(I)$, where $f^{(n)}(I)$ denotes the nth derivative of $f(I)$.)

8.5.2 Consider the driven pendulum $\phi'' + \alpha\phi' + \sin\phi = I$. By numerical computation of the phase portrait, verify that if α is fixed and sufficiently small, the system's stable limit cycle is destroyed in a homoclinic bifurcation as I decreases. Show that if α is too large, the bifurcation is an infinite-period bifurcation instead.

8.5.3 (Logistic equation with periodically varying carrying capacity) Consider the logistic equation $\dot{N} = rN(1 - N/K(t))$, where the carrying capacity is positive, smooth, and T-periodic in t.

a) Using a Poincaré map argument like that in the text, show that the system has at least one stable limit cycle of period T, contained in the strip $K_{\min} \le N \le K_{\max}$.
b) Is the cycle necessarily unique?

8.5.4 (Logistic equation with sinusoidal harvesting) In Exercise 3.7.3 you were asked to consider a simple model of a fishery with constant harvesting. Now consider a generalization in which the harvesting varies periodically in time, perhaps due to daily or seasonal variations. To keep things simple, assume the periodic harvesting is purely sinusoidal (Benardete et al. 2008). Then, if the fish population grows logistically in the absence of harvesting, the model is given in dimensionless form by $\dot{x} = rx(1-x) - h(1+\alpha\sin t)$. Assume that $r, h > 0$ and $0 < \alpha < 1$.

a) Show that if $h > r/4$ the system has no periodic solutions, even though the fish are being harvested periodically with period $T = 2\pi$. What happens to the fish population in this case?
b) By using a Poincaré map argument like that in the text, show that if $h < \dfrac{r}{4(1+\alpha)}$, there exists a 2π-periodic solution—in fact, a stable limit cycle—in the strip $1/2 < x < 1$. Similarly, show there exists an unstable limit cycle in the strip $0 < x < 1/2$. Interpret your results biologically.
c) What happens in between cases (a) and (b), i.e., for $\dfrac{r}{4(1+\alpha)} < h < \dfrac{r}{4}$?

8.5.5 (Driven pendulum with quadratic damping) Consider a pendulum driven by a constant torque and damped by air resistance. In dimensionless form, the governing equation is

$$\ddot{\theta}+\alpha\dot{\theta}\left|\dot{\theta}\right|+\sin\theta=F$$

where $\alpha>0$ and $F>0$ are the dimensionless damping strength and torque, respectively. The new feature here is that we assume the damping is quadratic, rather than linear, in the velocity $v=\dot{\theta}$. This is more realistic if the damping is primarily due to drag, but the trouble is that the damping becomes nonlinear, which normally would make the analysis harder. But as you'll see, the pleasant surprise here is that quadratic damping actually makes the system easier—in fact, it becomes explicitly solvable!

a) Find and classify the fixed points in the (θ,v) phase plane. If you find a center according to the linearization, decide whether this borderline case is truly a nonlinear center, a stable spiral, or an unstable spiral. (Hint: Find a local Liapunov function in the neighborhood of the fixed point.)

b) For $F>1$, prove that the system has a stable limit cycle (where we now regard the phase space as a cylinder rather than a plane). Then prove that this limit cycle is unique.

   Remarkably, exact formulas for the limit cycle and the homoclinic bifurcation curve can be found (Pedersen and Saermark 1973); this is one of the advantages of the quadratically damped case. However, the solution involves some tricky changes of variables. Here's how it works:

c) In the region $v>0$, $\theta(t)$ increases monotonically. Therefore it can be inverted formally to yield $t(\theta)$. Now regard θ as a new independent (time-like) variable, and introduce the new dependent variable $u=\frac{1}{2}v^2$. Use the chain rule (carefully, showing all your steps) to deduce that $\dfrac{du}{d\theta}=\ddot{\theta}$.

d) Hence, in the region $v>0$, the pendulum equation becomes $\dfrac{du}{d\theta}+2\alpha u+\sin\theta=F$, which is a linear equation in $u(\theta)$. Assuming that this equation is correct (even if you were unable to derive it), find an exact formula for the limit cycle when $F>1$.

e) Now decrease F while keeping α fixed. Show that the limit cycle undergoes a homoclinic bifurcation at some critical value of F, call it $F=F_c(\alpha)$, and give an exact formula for the bifurcation curve $F_c(\alpha)$.

8.6 Coupled Oscillators and Quasiperiodicity

8.6.1 ("Oscillator death" and bifurcations on a torus) In a paper on systems of neural oscillators, Ermentrout and Kopell (1990) illustrated the notion of "oscillator death" with the following model:

$$\dot{\theta}_1 = \omega_1 + \sin\theta_1 \cos\theta_2, \qquad \dot{\theta}_2 = \omega_2 + \sin\theta_2 \cos\theta_1,$$

where $\omega_1, \omega_2 \geq 0$.

a) Sketch all the qualitatively different phase portraits that arise as ω_1, ω_2 vary.
b) Find the curves in ω_1, ω_2 parameter space along which bifurcations occur, and classify the various bifurcations.
c) Plot the stability diagram in ω_1, ω_2 parameter space.

8.6.2 Reconsider the system (8.6.1):

$$\dot{\theta}_1 = \omega_1 + K_1 \sin(\theta_2 - \theta_1), \qquad \dot{\theta}_2 = \omega_2 + K_2 \sin(\theta_1 - \theta_2).$$

a) Show that the system has no fixed points, given that $\omega_1, \omega_2 > 0$ and $K_1, K_2 > 0$.
b) Find a conserved quantity for the system. (Hint: Solve for $\sin(\theta_2 - \theta_1)$ in two ways. The existence of a conserved quantity shows that this system is a non-generic flow on the torus; normally there would not be any conserved quantities.)
c) Suppose that $K_1 = K_2$. Show that the system can be nondimensionalized to

$$d\theta_1/d\tau = 1 + a\sin(\theta_2 - \theta_1), \qquad d\theta_2/d\tau = \omega + a\sin(\theta_1 - \theta_2).$$

d) Find the *winding number* $\lim_{\tau\to\infty} \theta_1(\tau)/\theta_2(\tau)$ analytically. (Hint: Evaluate the long-time averages $\langle d(\theta_1 + \theta_2)/d\tau \rangle$ and $\langle d(\theta_1 - \theta_2)/d\tau \rangle$, where the brackets are defined by $\langle f \rangle \equiv \lim_{T\to\infty} \frac{1}{T}\int_0^T f(\tau)d\tau$. For another approach, see Guckenheimer and Holmes (1983, p. 299).)

8.6.3 (Irrational flow yields dense orbits) Consider the flow on the torus given by $\dot{\theta}_1 = \omega_1, \dot{\theta}_2 = \omega_2$, where ω_1/ω_2 is irrational. Show each trajectory is ***dense***; i.e., given any point p on the torus, any initial condition q, and any $\varepsilon > 0$, there is some $t < \infty$ such that the trajectory starting at q passes within a distance ε of p.

8.6.4 Consider the system

$$\dot{\theta}_1 = E - \sin\theta_1 + K\sin(\theta_2 - \theta_1), \qquad \dot{\theta}_2 = E + \sin\theta_2 + K\sin(\theta_1 - \theta_2)$$

where $E, K \geq 0$.

a) Find and classify all the fixed points.
b) Show that if E is large enough, the system has periodic solutions on the torus. What type of bifurcation creates the periodic solutions?

c) Find the bifurcation curve in (E, K) space at which these periodic solutions are created.

A generalization of this system to $N >> 1$ phases has been proposed as a model of switching in charge-density waves (Strogatz et al. 1988, 1989).

8.6.5 (Plotting Lissajous figures) Using a computer, plot the curve whose parametric equations are $x(t) = \sin t$, $y(t) = \sin \omega t$, for the following rational and irrational values of the parameter ω:

(a) $\omega = 3$ (b) $\omega = \frac{2}{3}$ (c) $\omega = \frac{5}{3}$

(d) $\omega = \sqrt{2}$ (e) $\omega = \pi$ (f) $\omega = \frac{1}{2}(1+\sqrt{5})$.

The resulting curves are called *Lissajous figures.* In the old days they were displayed on oscilloscopes by using two ac signals of different frequencies as inputs.

8.6.6 (Explaining Lissajous figures) Lissajous figures are one way to visualize the knots and quasiperiodicity discussed in the text. To see this, consider a pair of uncoupled harmonic oscillators described by the four-dimensional system $\ddot{x} + x = 0$, $\ddot{y} + \omega^2 y = 0$.

a) Show that if $x = A(t) \sin \theta(t)$, $y = B(t) \sin \phi(t)$, then $\dot{A} = \dot{B} = 0$ (so A, B are constants) and $\dot{\theta} = 1$, $\dot{\theta} = \omega$.
b) Explain why (a) implies that trajectories are typically confined to two-dimensional tori in a four-dimensional phase space.
c) How are the Lissajous figures related to the trajectories of this system?

8.6.7 (Mechanical example of quasiperiodicity) The equations

$$m\ddot{r} = \frac{h^2}{mr^3} - k, \quad \dot{\theta} = \frac{h}{mr^2}$$

govern the motion of a mass m subject to a central force of constant strength $k > 0$. Here r, θ are polar coordinates and $h > 0$ is a constant (the angular momentum of the particle).

a) Show that the system has a solution $r = r_0$, $\dot{\theta} = \omega_\theta$, corresponding to uniform circular motion at a radius r_0 and frequency ω_θ. Find formulas for r_0 and ω_θ.
b) Find the frequency ω_r of small radial oscillations about the circular orbit.
c) Show that these small radial oscillations correspond to quasiperiodic motion by calculating the winding number ω_r / ω_θ.
d) Show by a geometric argument that the motion is either periodic or quasiperiodic for *any* amplitude of radial oscillation. (To say it in a more interesting way, the motion is never chaotic.)
e) Can you think of a mechanical realization of this system?

8.6.8 Solve the equations of Exercise 8.6.7 on a computer, and plot the particle's path in the plane with polar coordinates r, θ.

8.6.9 (Japanese tree frogs) Many thanks to Bard Ermentrout for suggesting the following exercise. An isolated male Japanese tree frog will call nearly periodically. When two frogs are placed close together (say, 50 cm apart), they can hear each other calling and tend to adjust their croak rhythms so that they call in alternation, half a cycle apart—a form of phase-locking known as antiphase synchronization.

So what happens when three frogs interact? This situation frustrates them; there's no way all three can get half a cycle away from everyone else. Aihara et al. (2011) found experimentally that in this case, the three frogs settle into one of two distinctive patterns (and they occasionally seem to switch between them, probably due to noise in the environment). One stable pattern involves a pair of frogs calling in unison, with the third frog calling approximately half a cycle out of phase from both of them. The other stable pattern has the three frogs maximally out of sync, with each calling one-third of a cycle apart from the other two.

Aihara et al. (2011) explored a coupled oscillator model of these phenomena, the essence of which is contained in the following systems for two frogs,

$$\begin{aligned}\dot{\theta}_1 &= \omega + H(\theta_2 - \theta_1)\\ \dot{\theta}_2 &= \omega + H(\theta_1 - \theta_2),\end{aligned}$$

and three frogs,

$$\begin{aligned}\dot{\theta}_1 &= \omega + H(\theta_2 - \theta_1) + H(\theta_3 - \theta_1)\\ \dot{\theta}_2 &= \omega + H(\theta_1 - \theta_2) + H(\theta_3 - \theta_2)\\ \dot{\theta}_3 &= \omega + H(\theta_1 - \theta_3) + H(\theta_2 - \theta_3).\end{aligned}$$

Here θ_i denotes the phase of the calling rhythm of frog *i*, and the function H quantifies the interaction between any two of them. For simplicity we'll assume all the frogs are identically coupled (same H for all of them) and have identical natural frequencies ω. Furthermore, assume that H is odd, smooth, and 2π-periodic.

a) Rewrite the systems for both two and three frogs in terms of the phase differences $\phi = \theta_1 - \theta_2$ and $\psi = \theta_2 - \theta_3$.
b) Show that the experimental results for two frogs are consistent with the simplest possible interaction function, $H(x) = a\sin x$, if the sign of a is chosen appropriately. But then show that this simple H cannot account for the three-frog results.
c) Next, consider more complicated interaction functions of the form $H(x) = a\sin x + b\sin 2x$. For the three-frog model, use a computer to plot the phase portraits in the (ϕ, ψ) plane for various values of a and b. Show that for suitable choices of a and b, you can explain all the experimental results for two and three frogs. That is, you can find a domain in the (a, b) parameter space for which the system has:

i) a stable antiphase solution for the two-frog model;

ii) a stable phase-locked solution for the three-frog model, in which frogs 1 and 2 are in sync and approximately π out of phase from frog 3;

iii) a co-existing stable phase-locked solution with the three frogs one-third of a cycle apart.

d) Show numerically that adding a small even periodic component to H does not alter these results qualitatively.

Caveat: The three-frog model studied here is more symmetrical than that considered by Aihara et al. (2011). They assumed unequal coupling strengths because in their experiments one frog was positioned midway between the other two. The frogs at either end therefore interacted less strongly with each other than with the frog in the middle.

8.7 Poincaré Maps

8.7.1 Use partial fractions to evaluate the integral $\int_{r_0}^{r_1} \frac{dr}{r(1-r^2)}$ that arises in Example 8.7.1, and show that $r_1 = [1 + e^{-4\pi}(r_0^{-2} - 1)]^{-1/2}$. Then confirm that $P'(r^*) = e^{-4\pi}$, as expected from Example 8.7.3.

8.7.2 Consider the vector field on the cylinder given by $\dot{\theta} = 1,\ \ \dot{y} = ay$. Define an appropriate Poincaré map and find a formula for it. Show that the system has a periodic orbit. Classify its stability for all real values of a.

8.7.3 (Overdamped system forced by a square wave) Consider an overdamped linear oscillator (or an RC-circuit) forced by a square wave. The system can be nondimensionalized to $\dot{x} + x = F(t)$, where $F(t)$ is a square wave of period T. To be more specific, suppose

$$F(t) = \begin{cases} +A, & 0 < t < T/2 \\ -A, & T/2 < t < T \end{cases}$$

for $t \in (0, T)$, and then $F(t)$ is periodically repeated for all other t. The goal is to show that all trajectories of the system approach a unique periodic solution. We could try to solve for $x(t)$ but that gets a little messy. Here's an approach based on the Poincaré map—the idea is to "strobe" the system once per cycle.

a) Let $x(0) = x_0$. Show that $x(T) = x_0 e^{-T} - A(1 - e^{-T/2})^2$.

b) Show that the system has a unique periodic solution, and that it satisfies $x_0 = -A \tanh(T/4)$.

c) Interpret the limits of $x(T)$ as $T \to 0$ and $T \to \infty$. Explain why they're plausible.

d) Let $x_1 = x(T)$, and define the Poincaré map P by $x_1 = P(x_0)$. More generally, $x_{n+1} = P(x_n)$. Plot the graph of P.
e) Using a cobweb picture, show that P has a globally stable fixed point. (Hence the original system eventually settles into a periodic response to the forcing.)

8.7.4 A Poincaré map for the system $\dot{x} + x = A \sin \omega t$ was shown in Figure 8.7.3, for a particular choice of parameters. Given that $\omega > 0$, can you deduce the sign of A? If not, explain why not.

8.7.5 (Another driven overdamped system) By considering an appropriate Poincaré map, prove that the system $\dot{\theta} + \sin\theta = \sin t$ has at least two periodic solutions. Can you say anything about their stability? (Hint: Regard the system as a vector field on a cylinder: $\dot{t} = 1$, $\dot{\theta} = \sin t - \sin\theta$. Sketch the nullclines and thereby infer the shape of certain key trajectories that can be used to bound the periodic solutions. For instance, sketch the trajectory that passes through $(t, \theta) = (\frac{\pi}{2}, \frac{\pi}{2})$.)

8.7.6 Give a mechanical interpretation of the system $\dot{\theta} + \sin\theta = \sin t$ considered in the previous exercise.

8.7.7 (Computer work) Plot a computer-generated phase portrait of the system $\dot{t} = 1$, $\dot{\theta} = \sin t - \sin\theta$. Check that your results agree with your answer to Exercise 8.7.5.

8.7.8 Consider the system $\dot{x} + x = F(t)$, where $F(t)$ is a smooth, T-periodic function. Is it true that the system necessarily has a stable T-periodic solution $x(t)$? If so, prove it; if not, find an F that provides a counterexample.

8.7.9 Consider the vector field given in polar coordinates by $\dot{r} = r - r^2$, $\dot{\theta} = 1$.
a) Compute the Poincaré map from S to itself, where S is the positive x-axis.
b) Show that the system has a unique periodic orbit and classify its stability.
c) Find the characteristic multiplier for the periodic orbit.

8.7.10 Explain how to find Floquet multipliers numerically, starting from perturbations along the coordinate directions.

8.7.11 (Reversibility and the in-phase periodic state of a Josephson array) Use a reversibility argument to prove that the in-phase periodic state of (8.7.1) is not attracting, even if the nonlinear terms are kept.

8.7.12 (Globally coupled oscillators) Consider the following system of N identical oscillators:

$$\dot{\theta}_i = f(\theta_i) + \frac{K}{N}\sum_{j=1}^{N} f(\theta_j), \quad \text{for } i = 1, \ldots, N,$$

where $K > 0$ and $f(\theta)$ is smooth and 2π-periodic. Assume that $f(\theta) > 0$ for all θ so that the in-phase solution is periodic. By calculating the linearized Poincaré map as in Example 8.7.4, show that all the characteristic multipliers equal $+1$.

Thus the neutral stability found in Example 8.7.4 holds for a broader class of oscillator arrays. In particular, the reversibility of the system is not essential. This example is from Tsang et al. (1991).